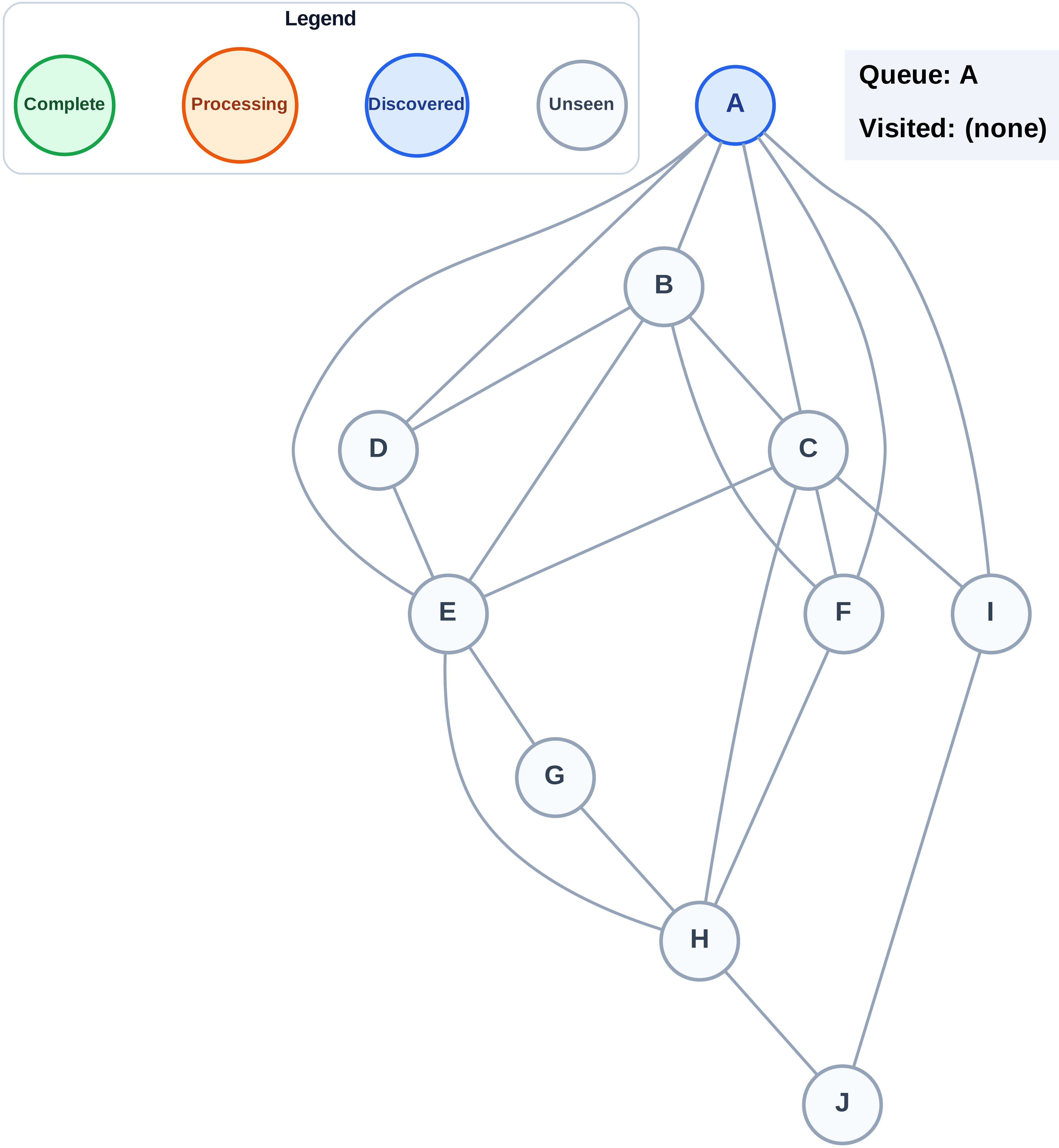
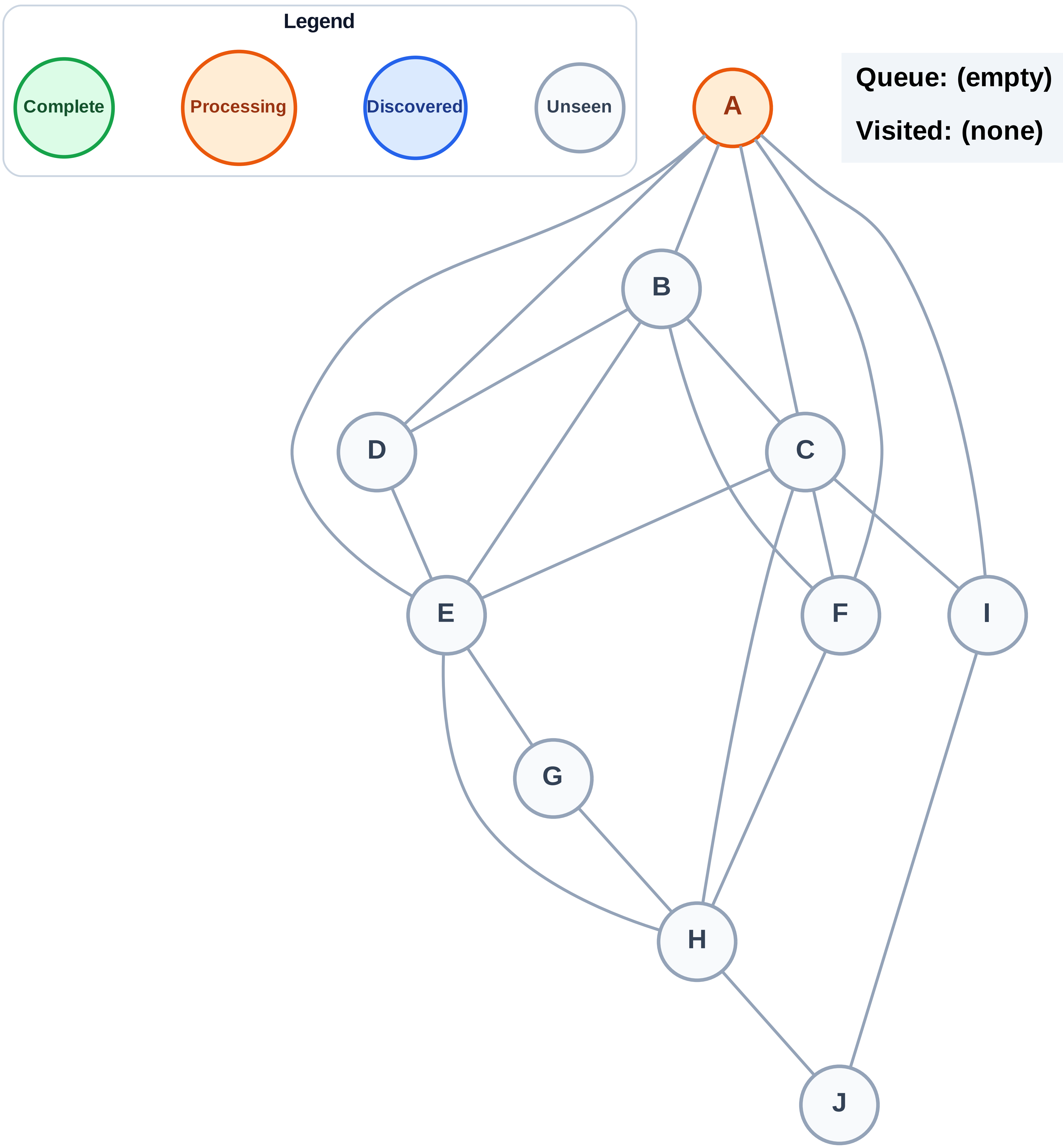
BFS Traversal

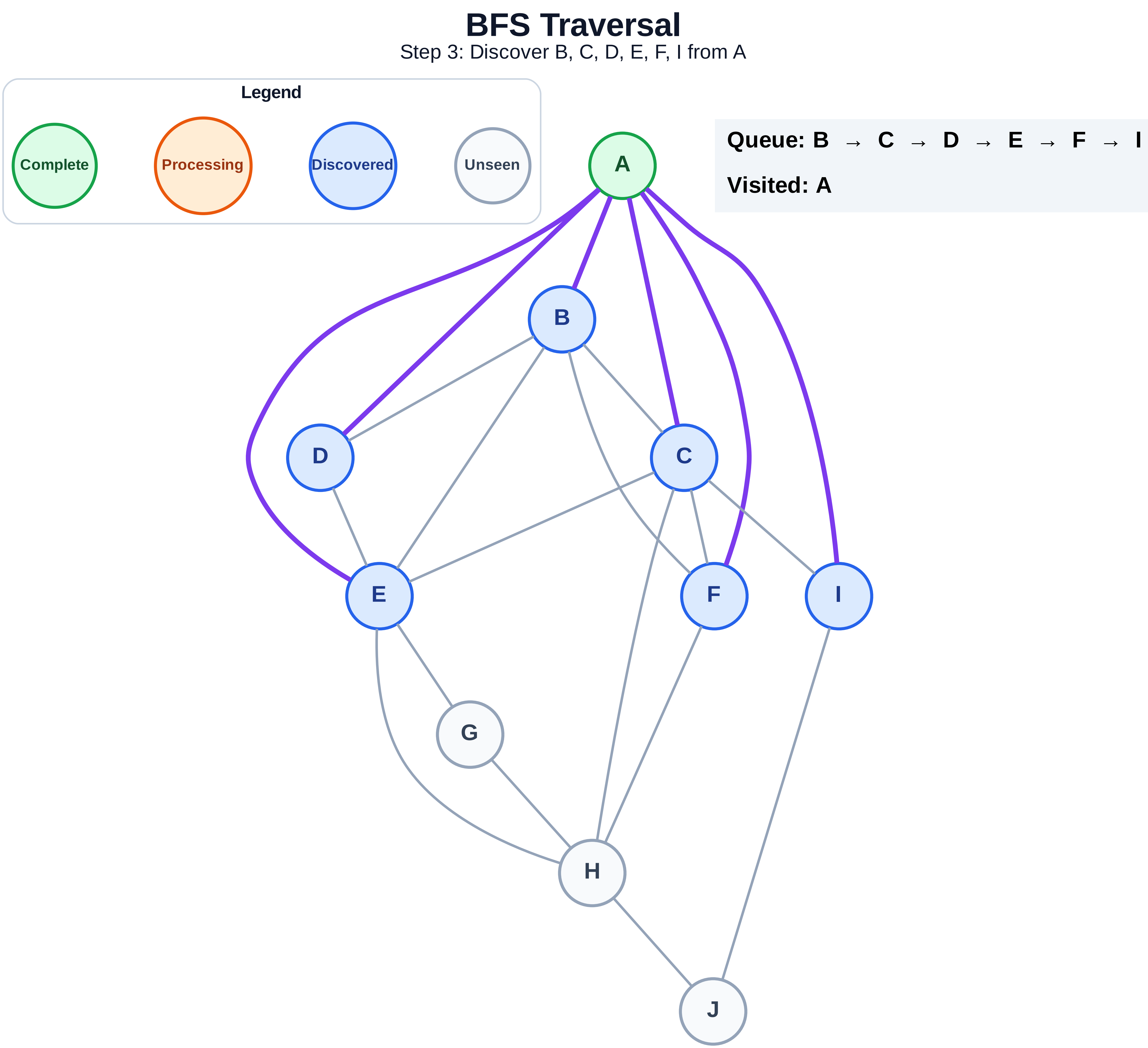
Step 1: Start at A



BFS Traversal

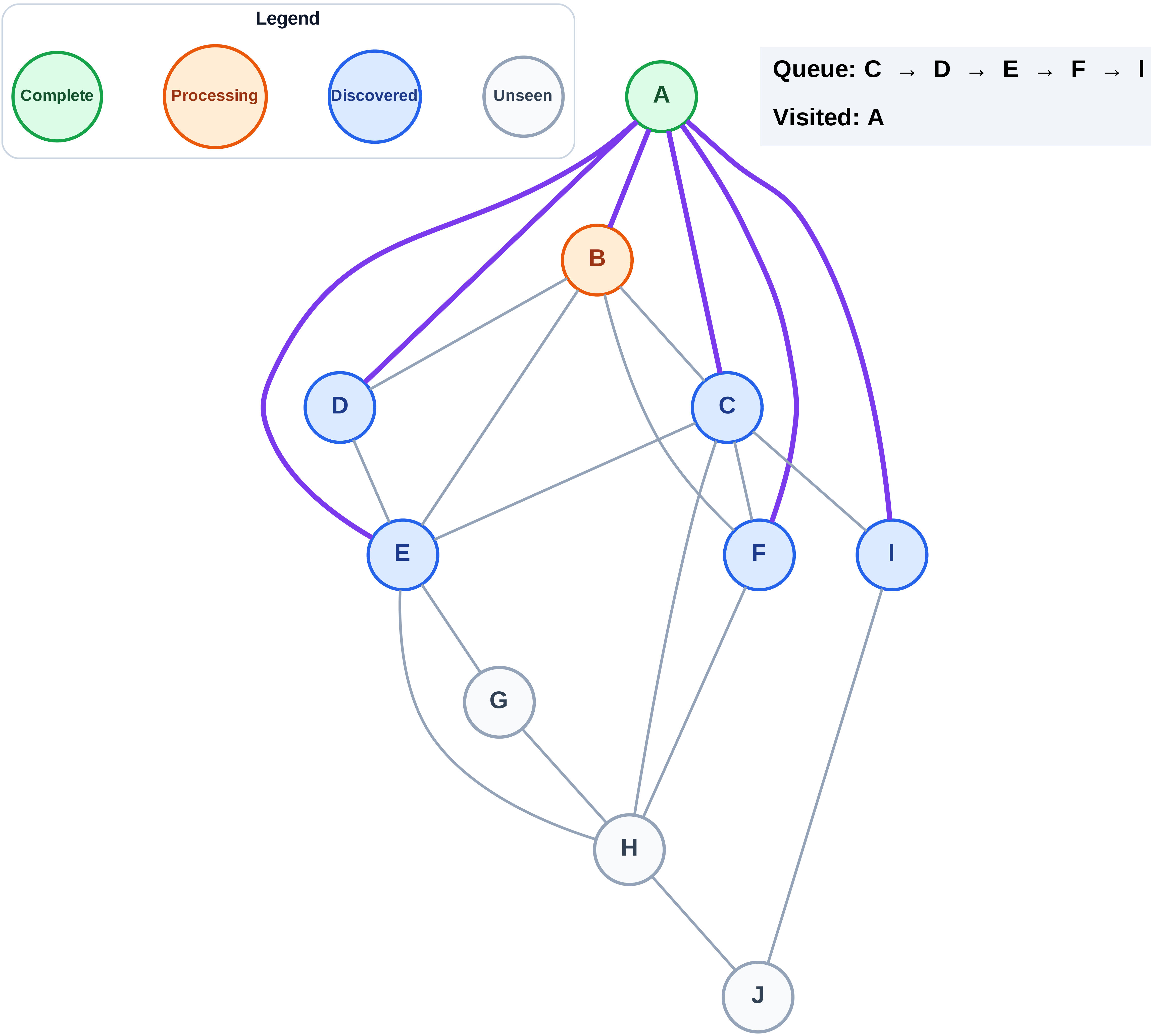
Step 2: Process node A




BFS Traversal
Step 3: Discover B, C, D, E, F, I from A
Legend
Complete
Processing
Discovered
Unseen
Queue: B → C → D → E → F → I
Visited: A
A
B
C
D
E
F
G
H
I
J

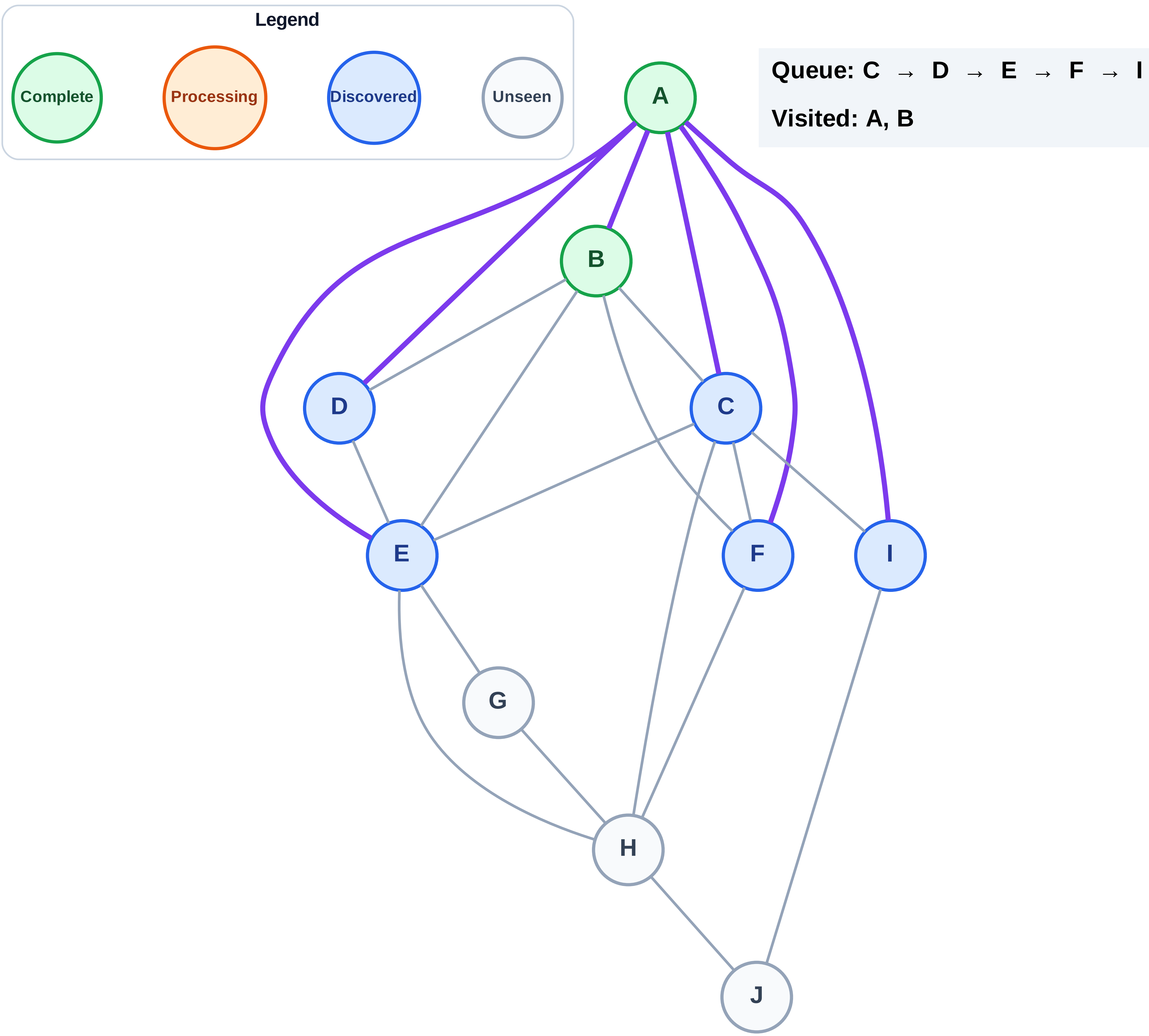
BFS Traversal

Step 4: Process node B



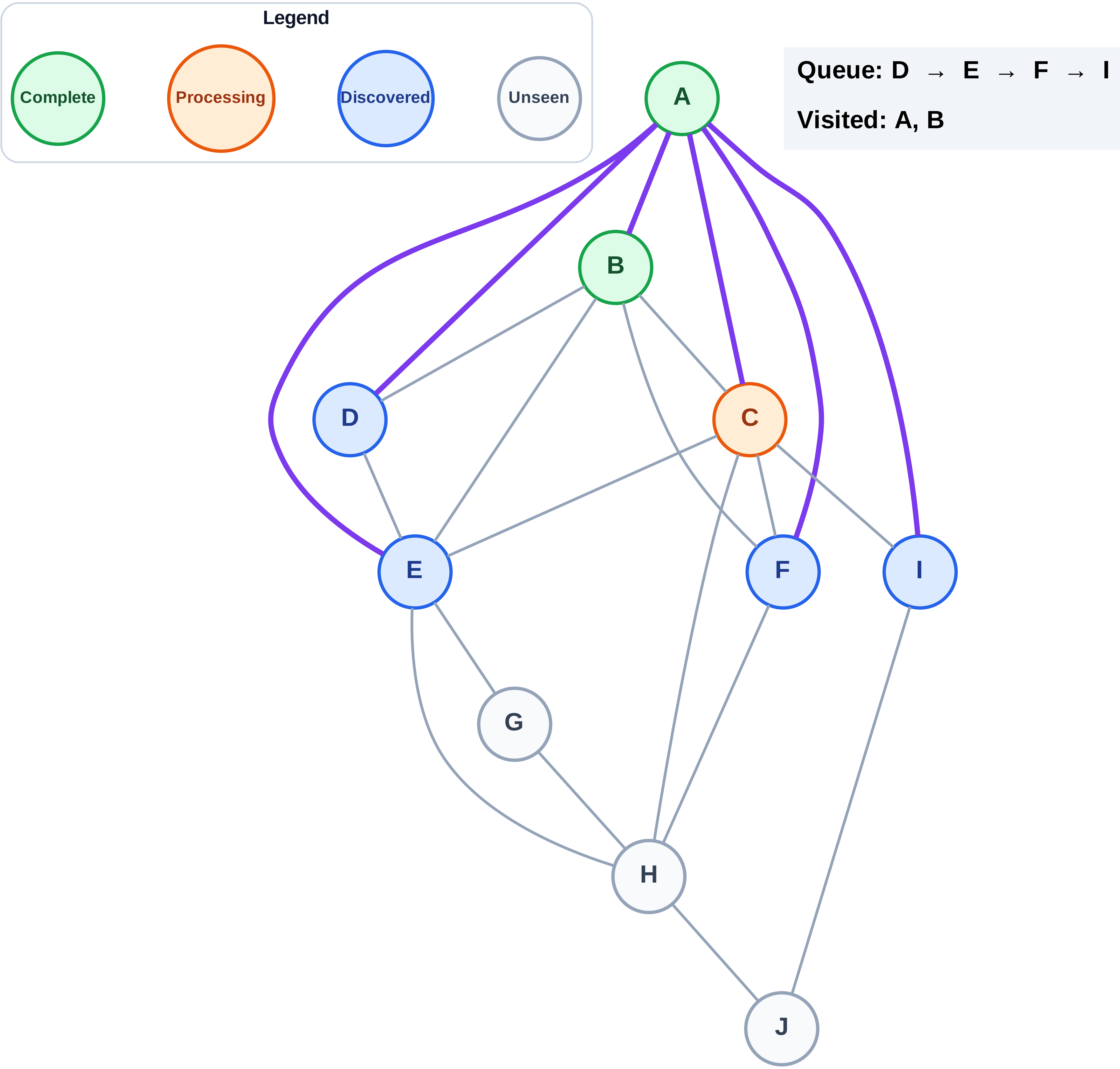
BFS Traversal

Step 5: No new neighbors from B



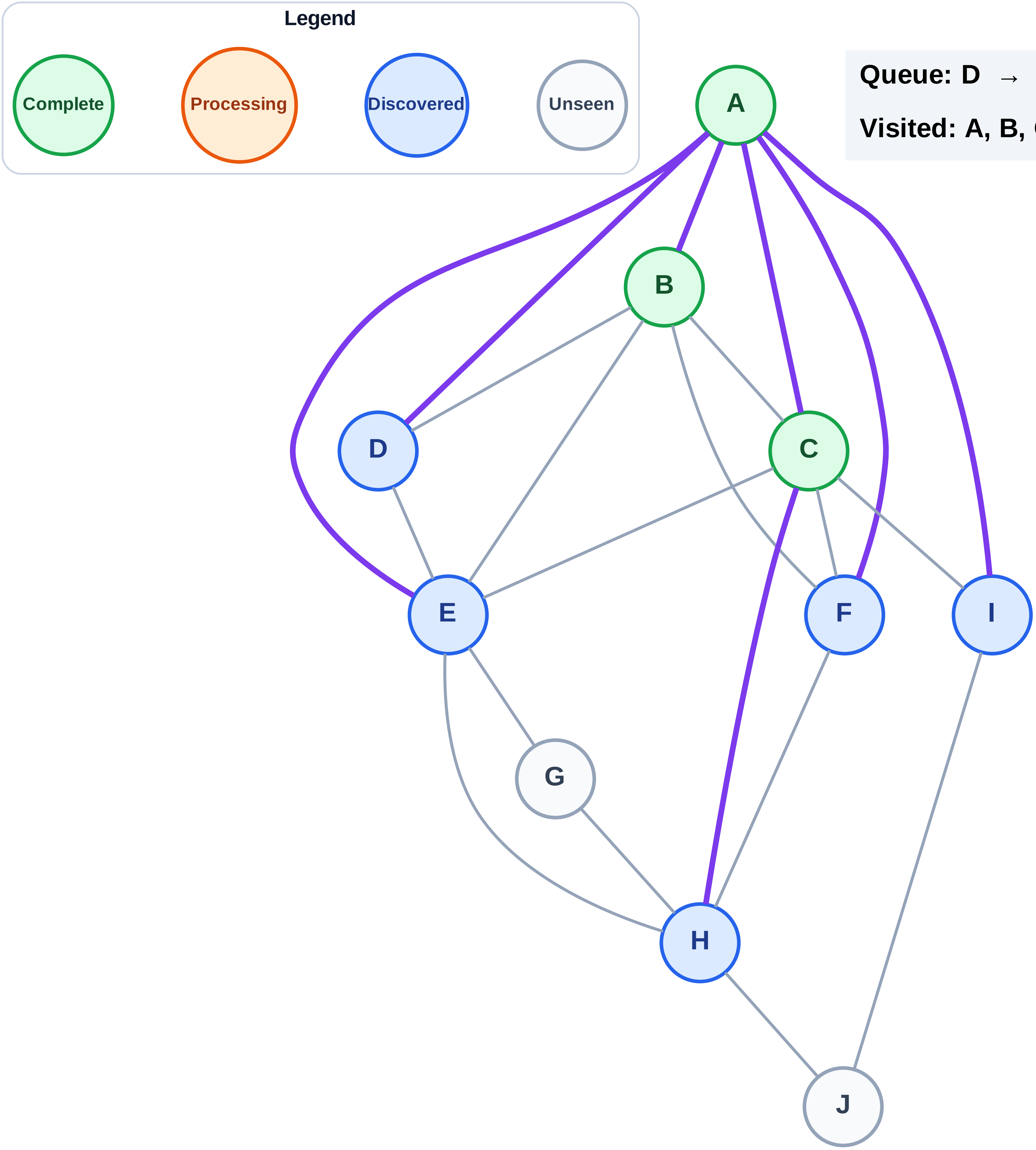
BFS Traversal

Step 6: Process node C



BFS Traversal

Step 7: Discover H from C

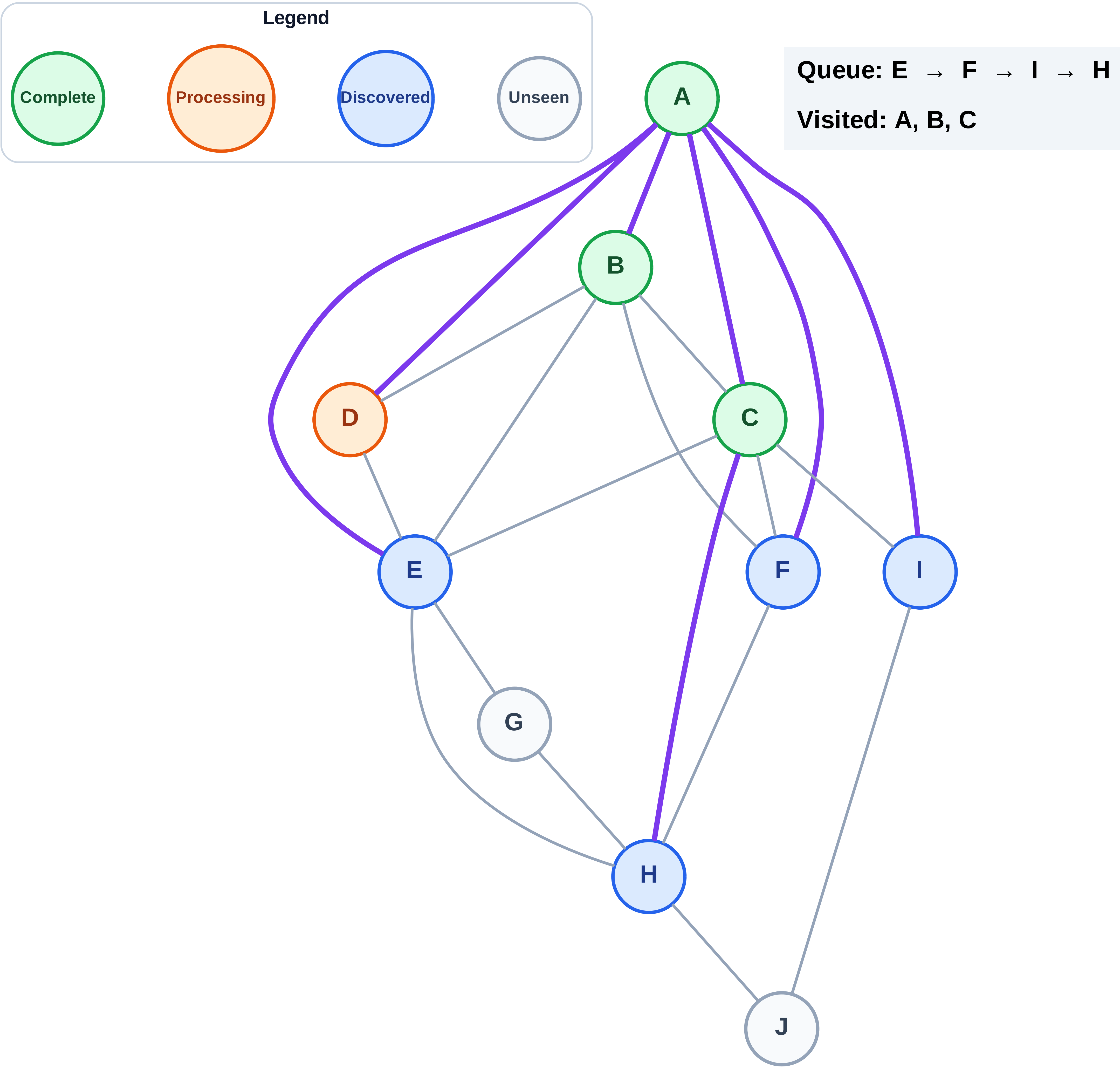


Queue: D → E → F → I → H

Visited: A, B, C

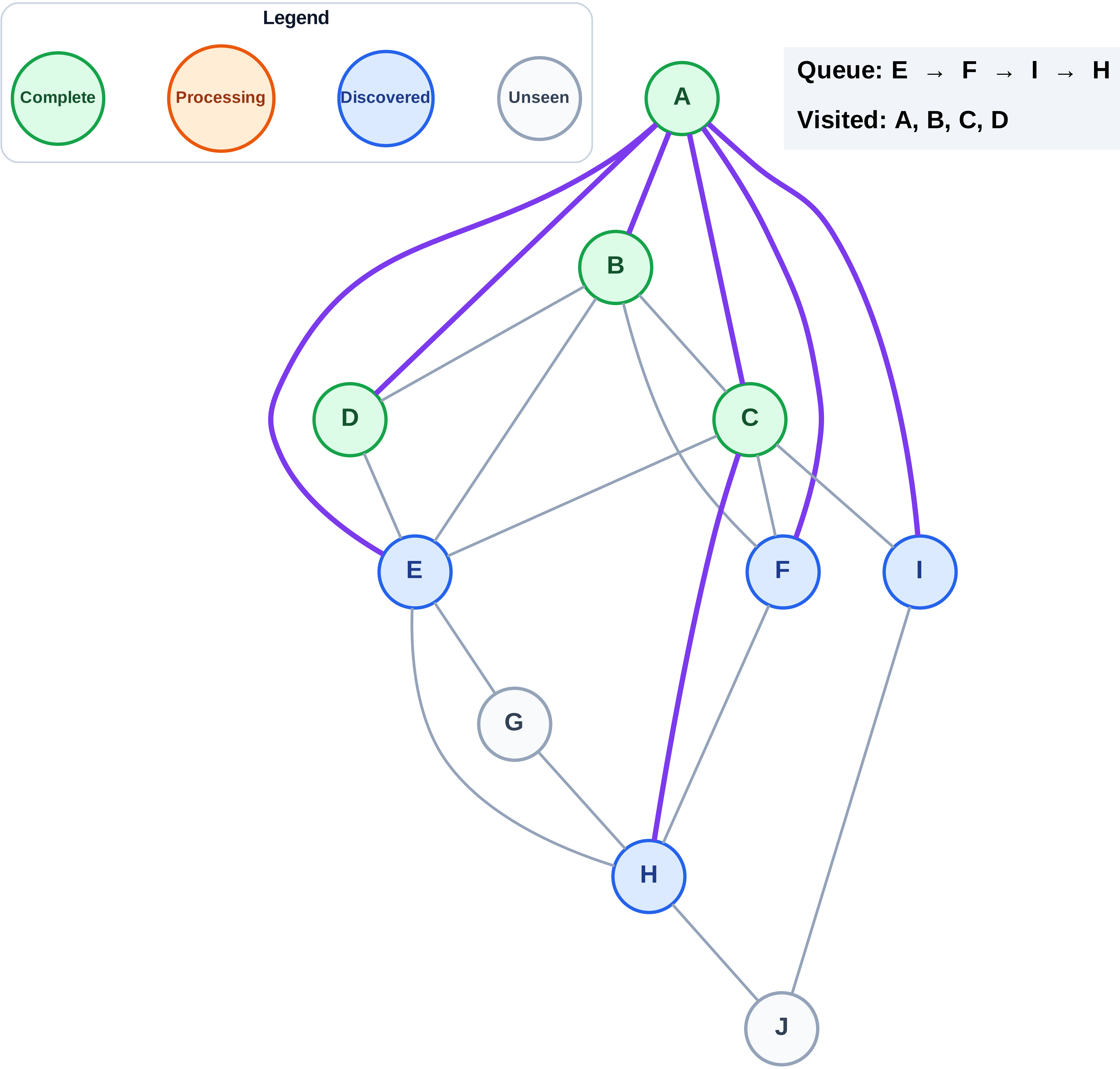
BFS Traversal

Step 8: Process node D



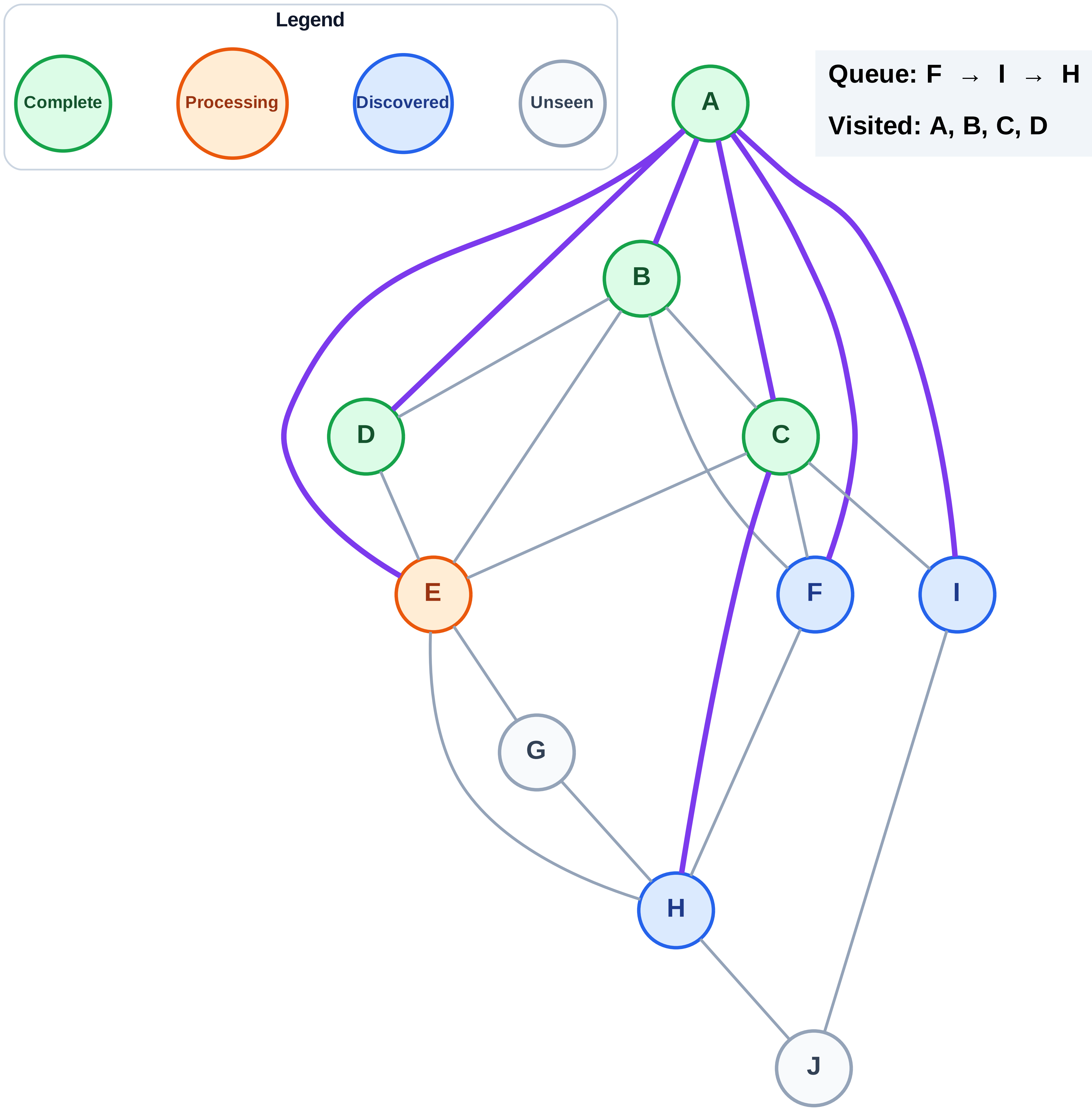
BFS Traversal

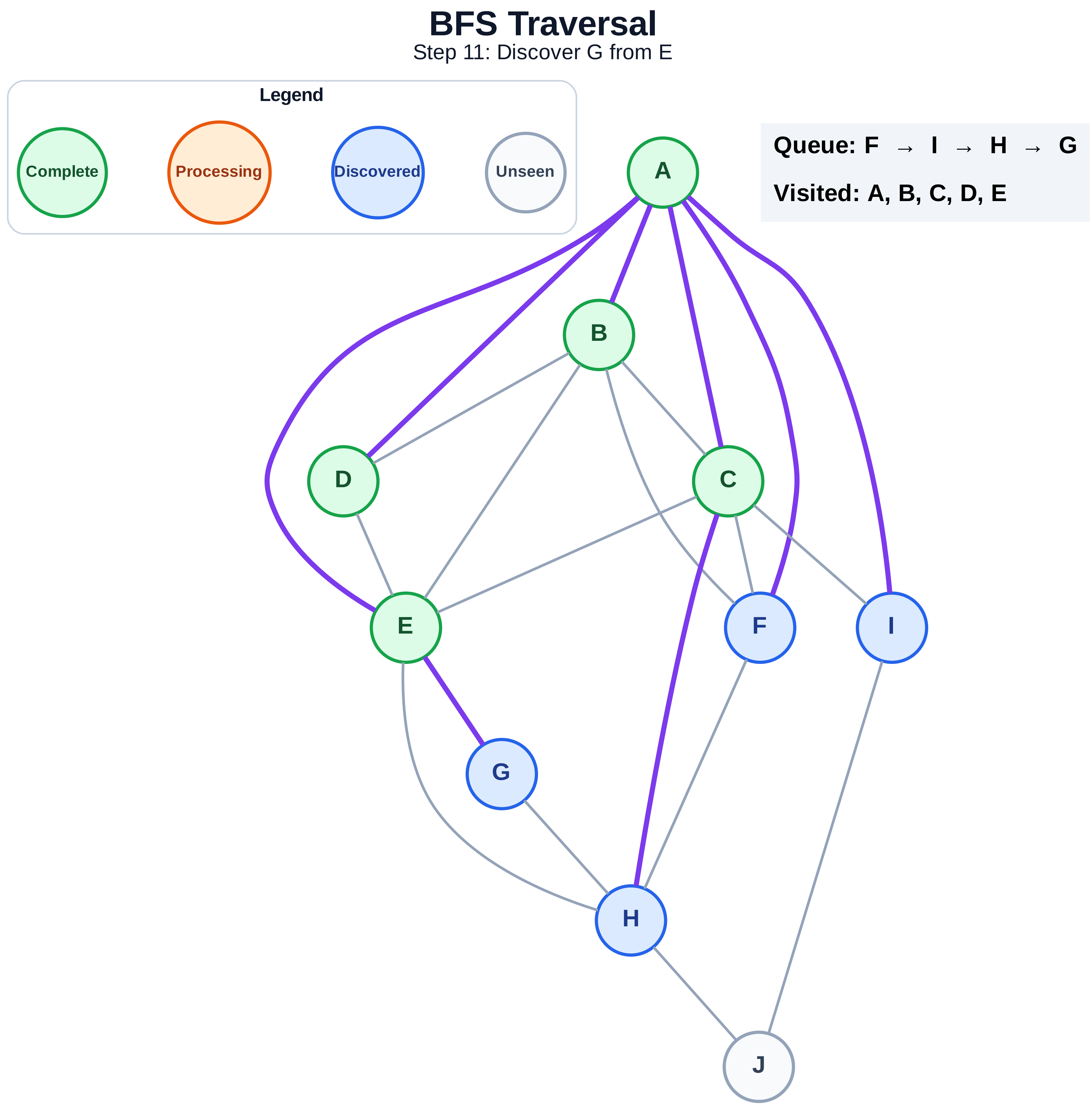
Step 9: No new neighbors from D

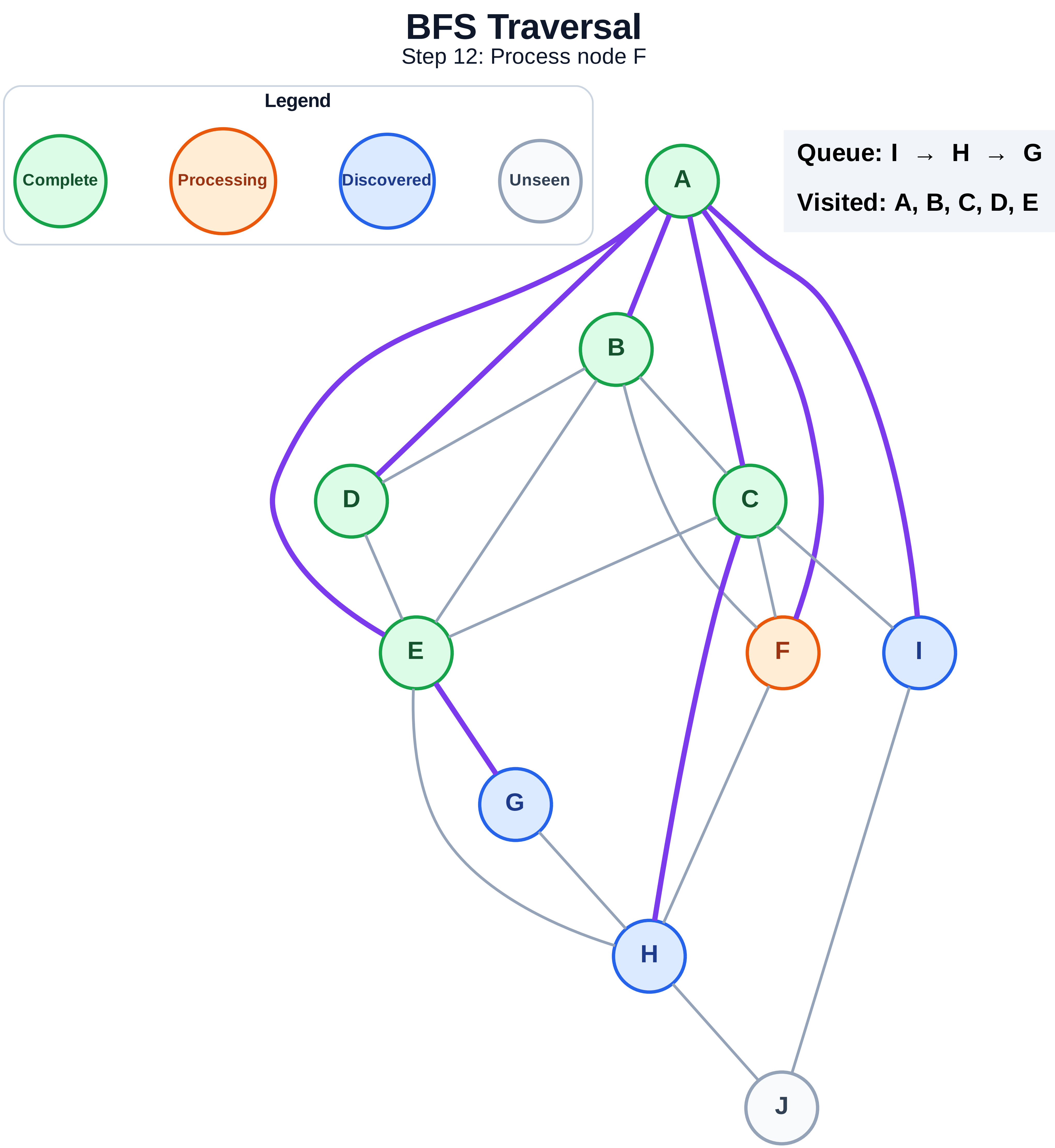


BFS Traversal

Step 10: Process node E



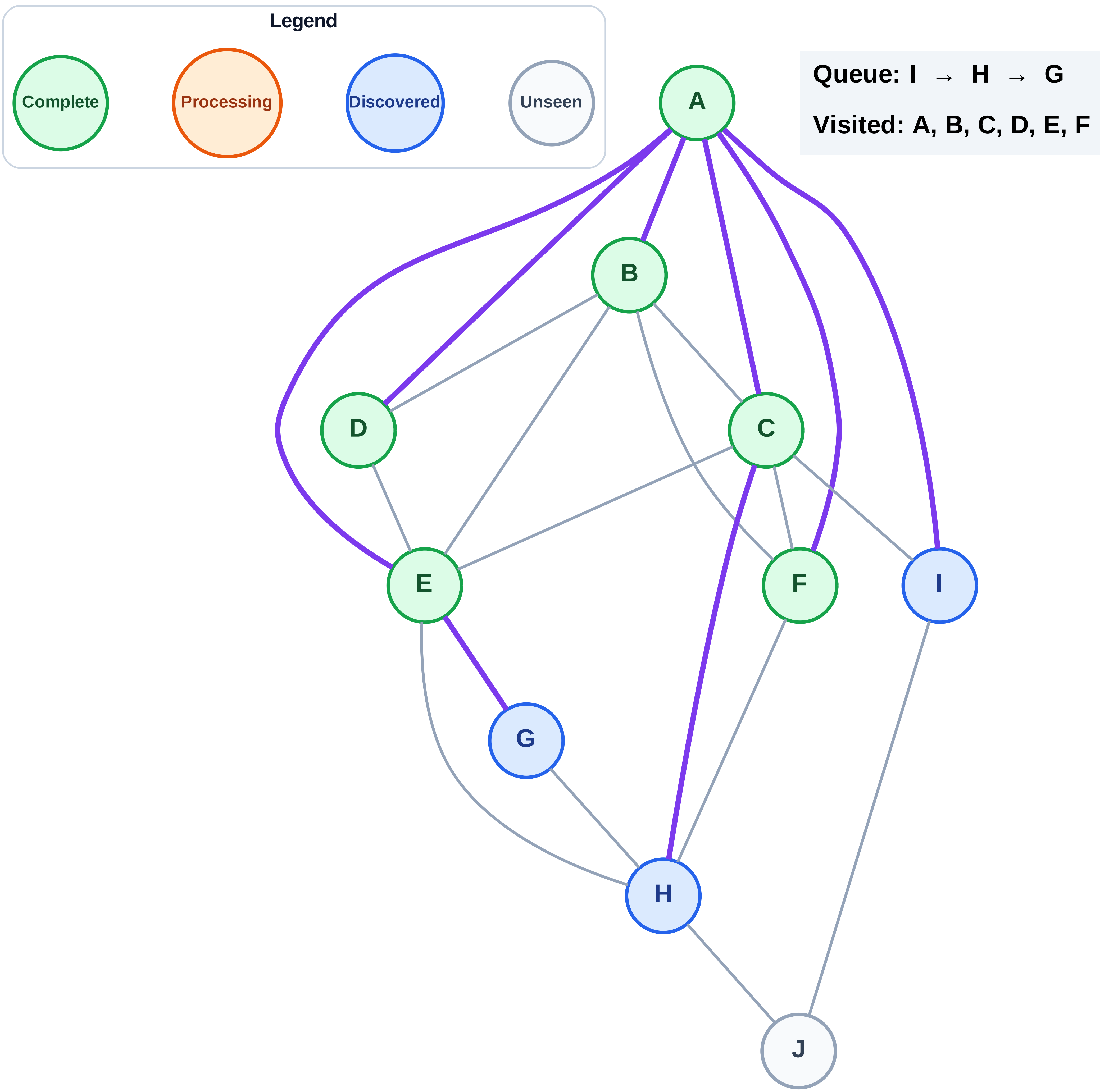

BFS Traversal
Step 11: Discover G from E
Legend
Complete
Processing
Discovered
Unseen
Queue: F → I → H → G
Visited: A, B, C, D, E
A
B
C
D
E
F
I
G
H
J



BFS Traversal
Step 12: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: I → H → G
Visited: A, B, C, D, E
A
B
C
D
E
F
G
H
I
J

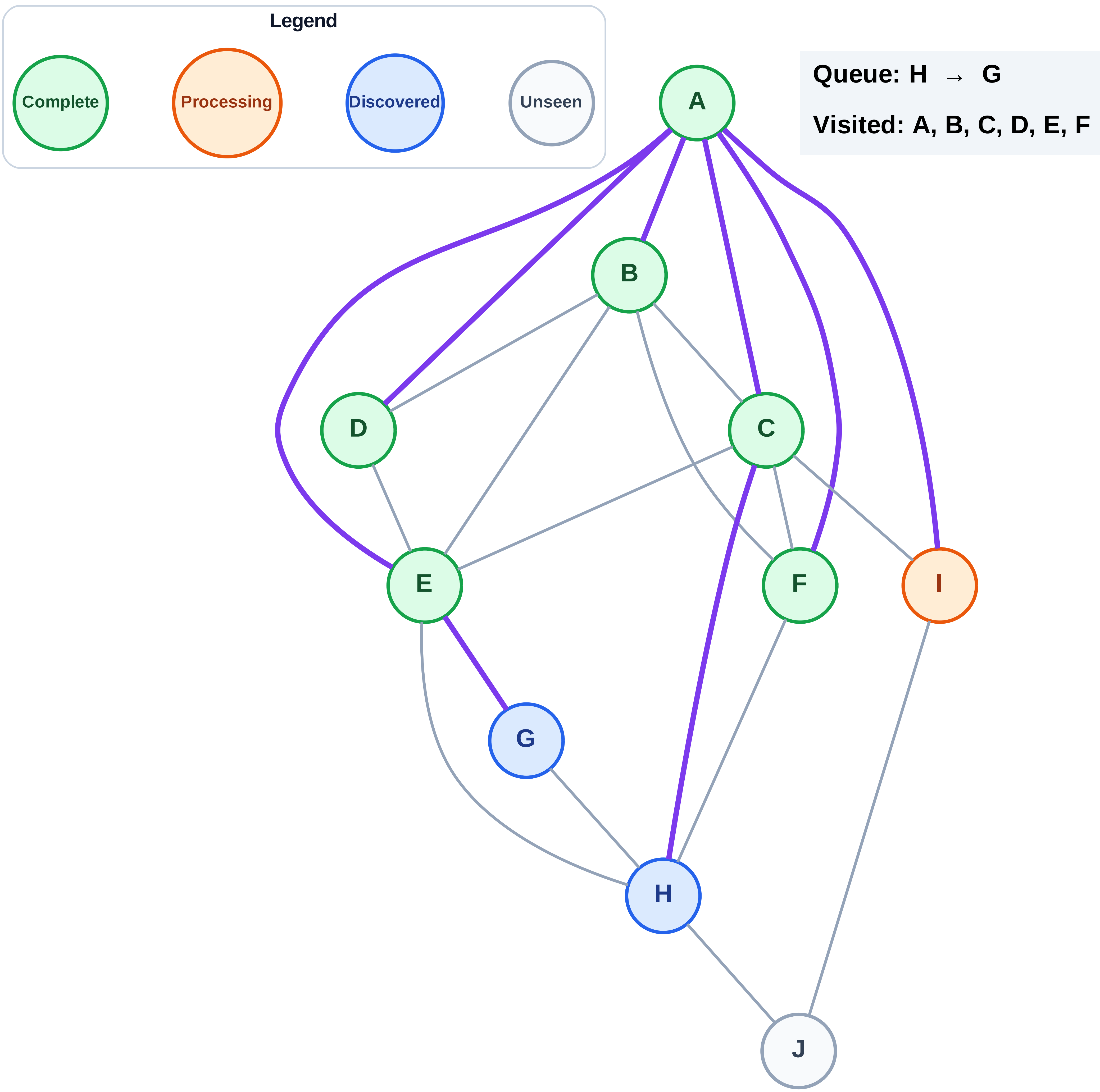
BFS Traversal

Step 13: No new neighbors from F



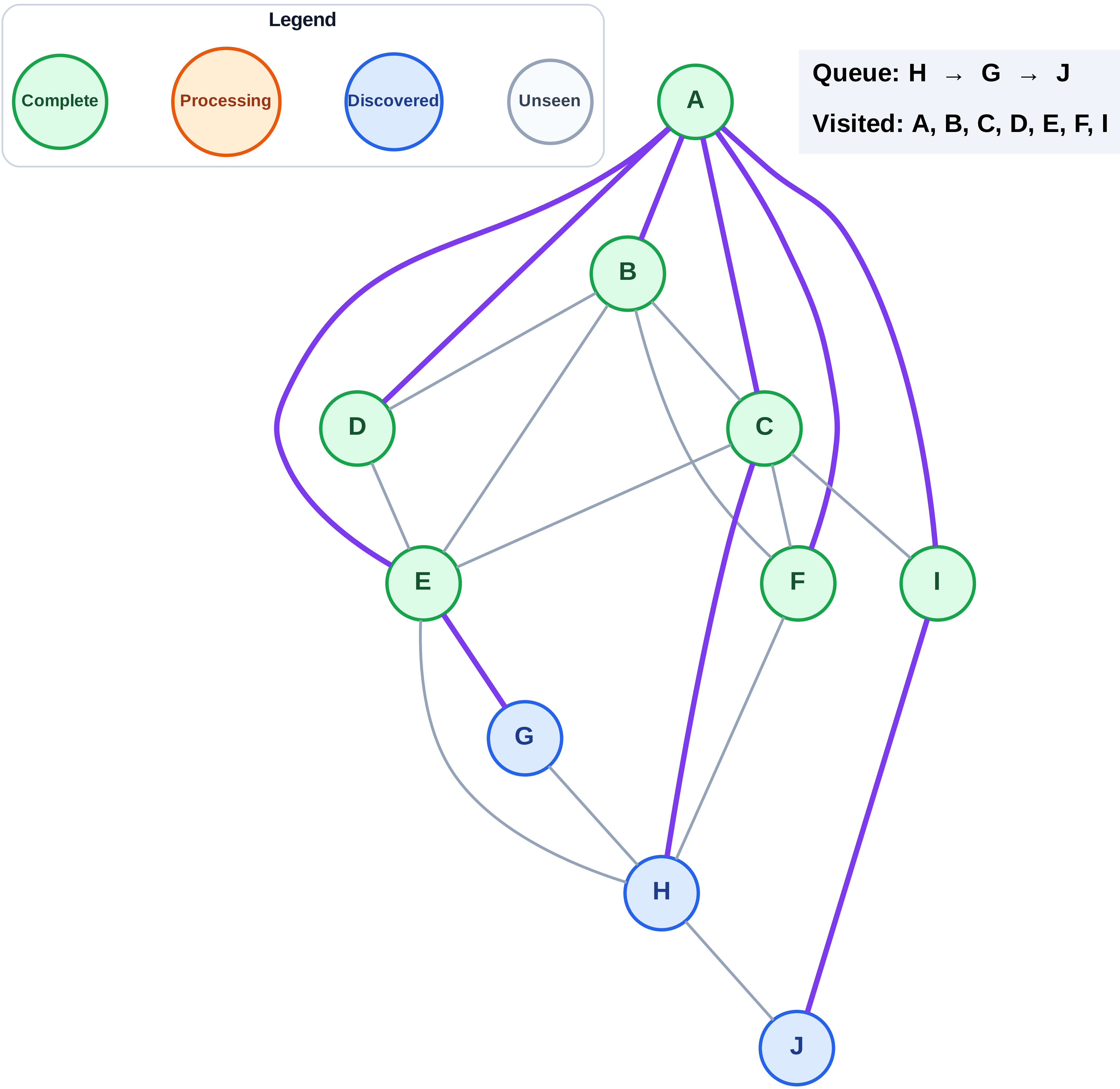
BFS Traversal

Step 14: Process node I



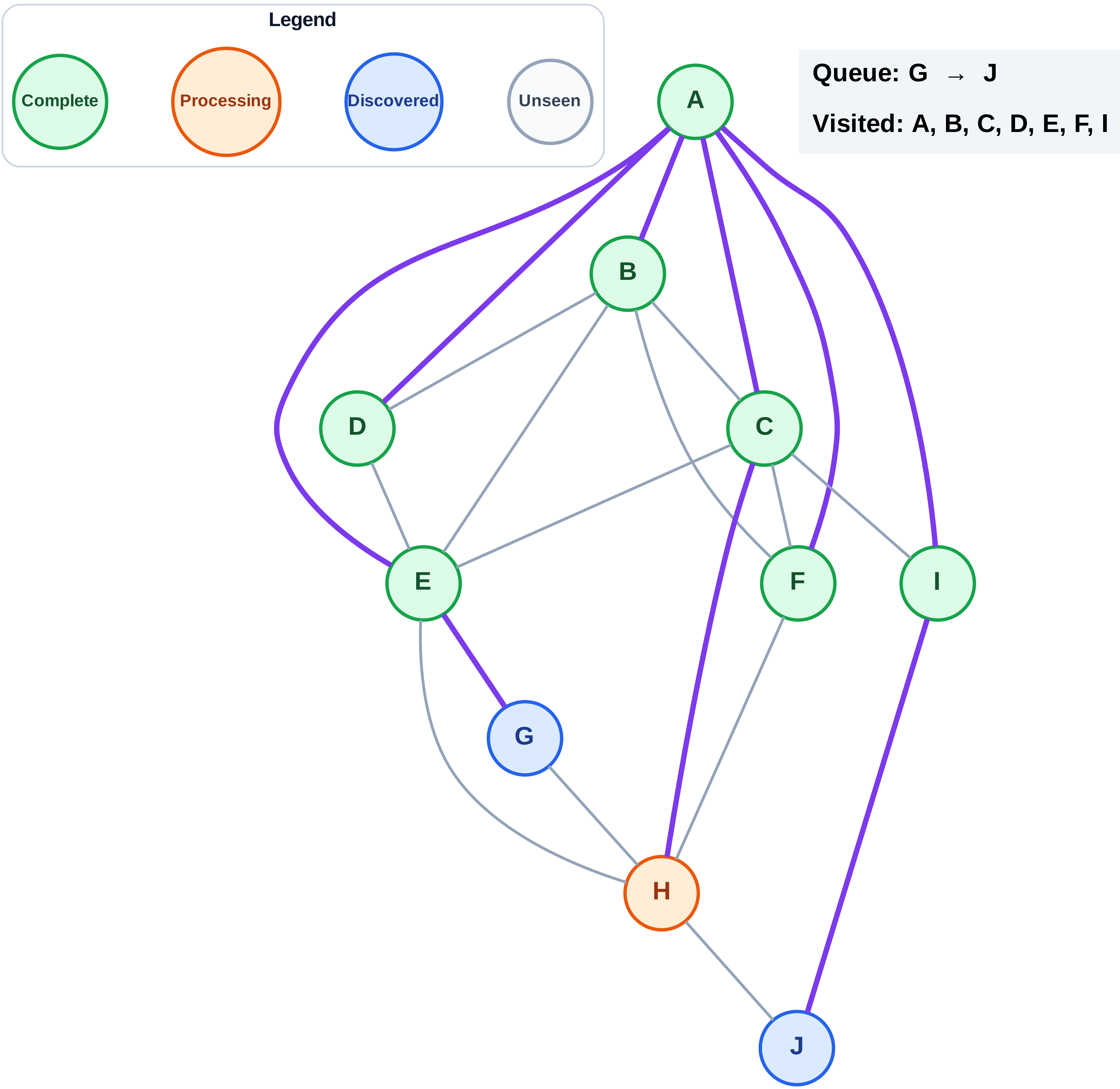
BFS Traversal

Step 15: Discover J from I



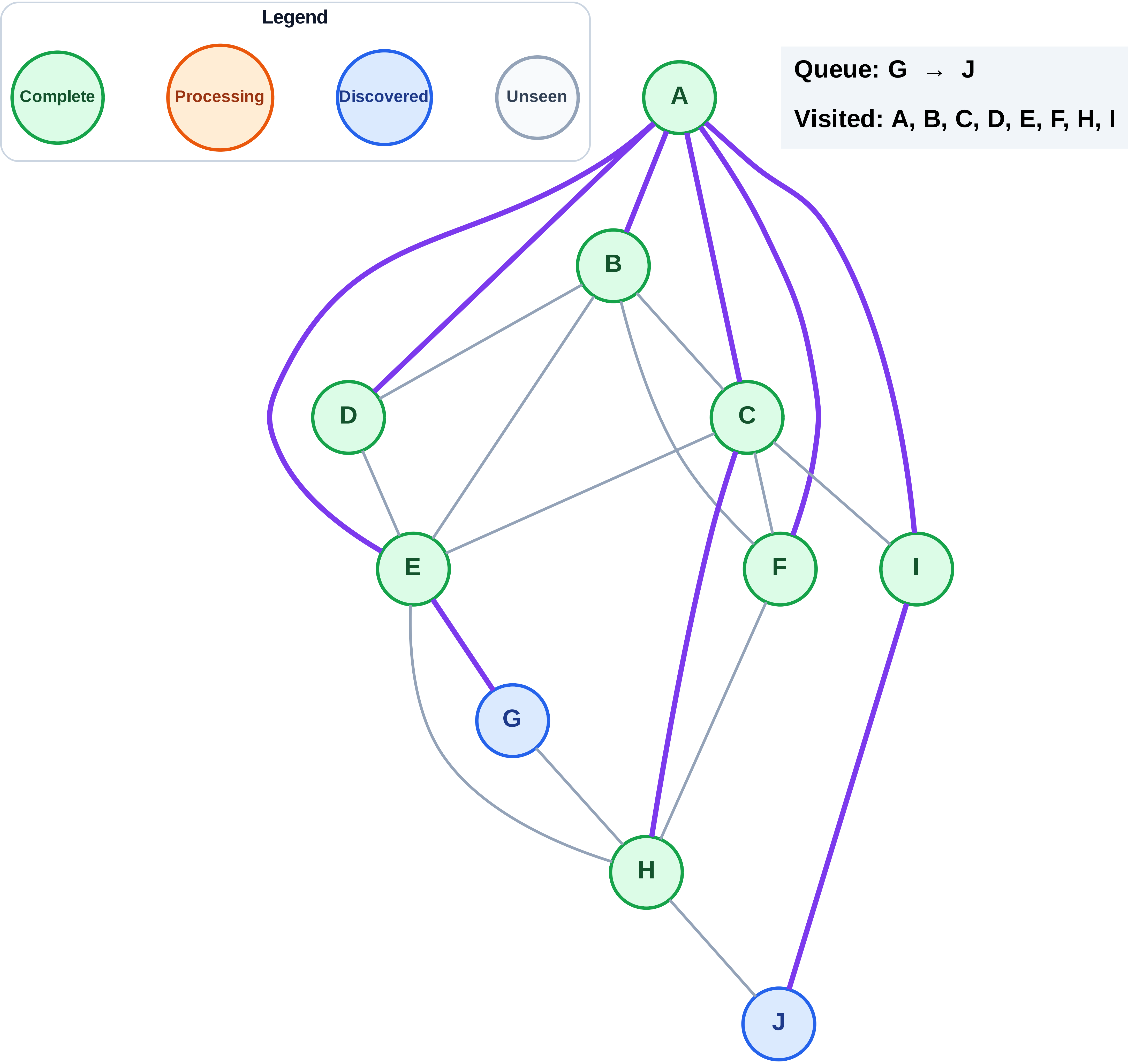
BFS Traversal

Step 16: Process node H



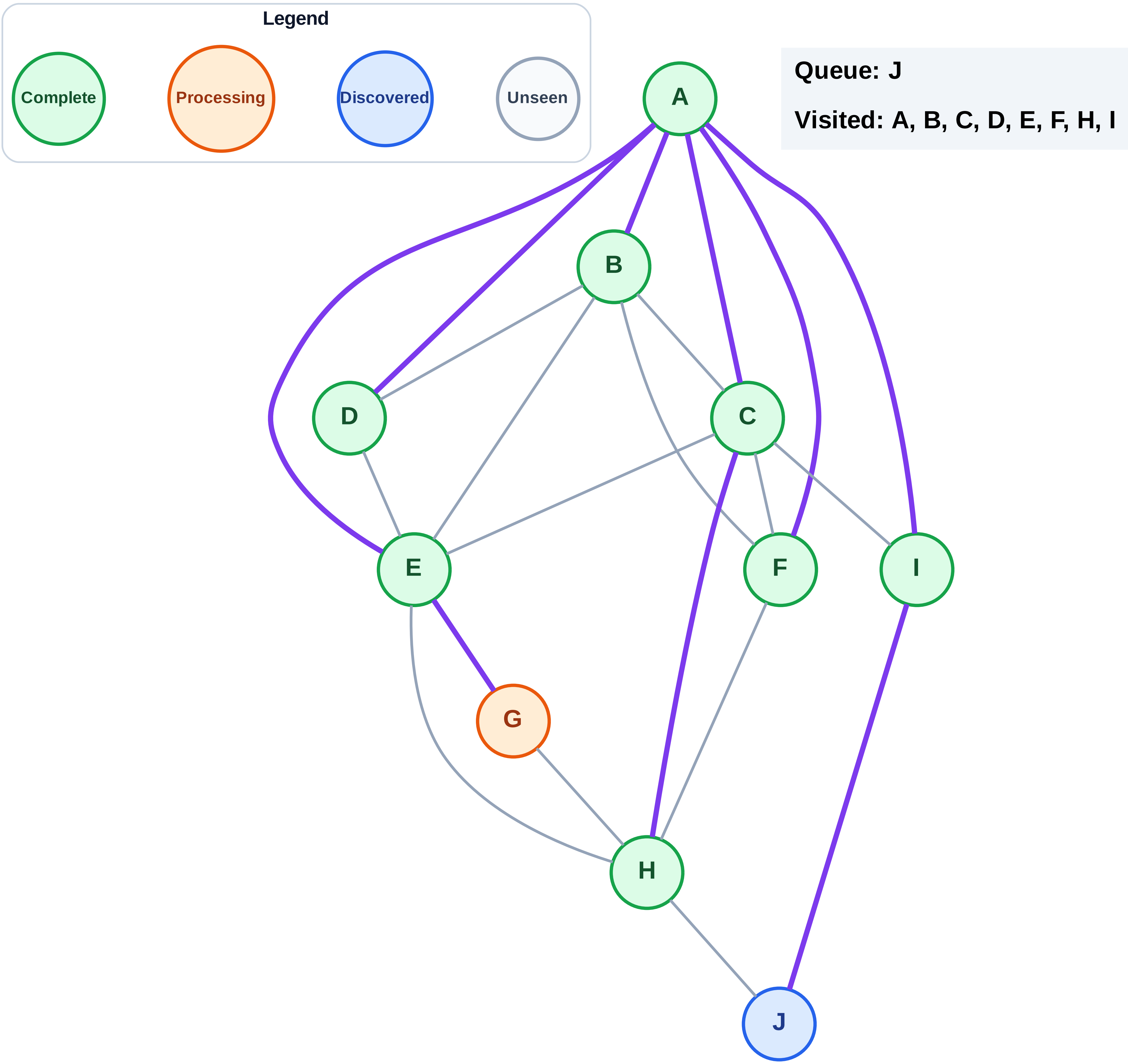
BFS Traversal

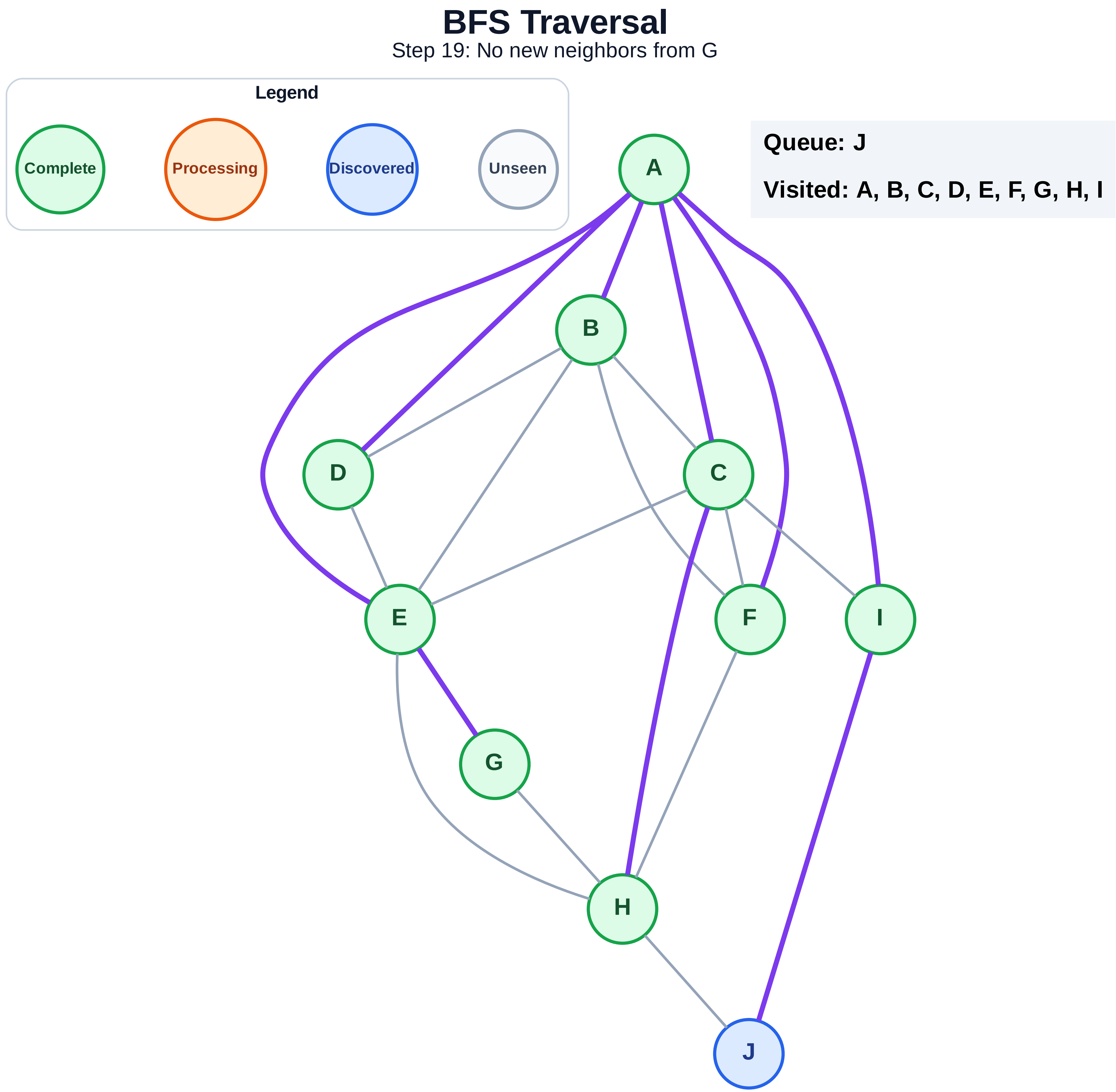
Step 17: No new neighbors from H



BFS Traversal

Step 18: Process node G

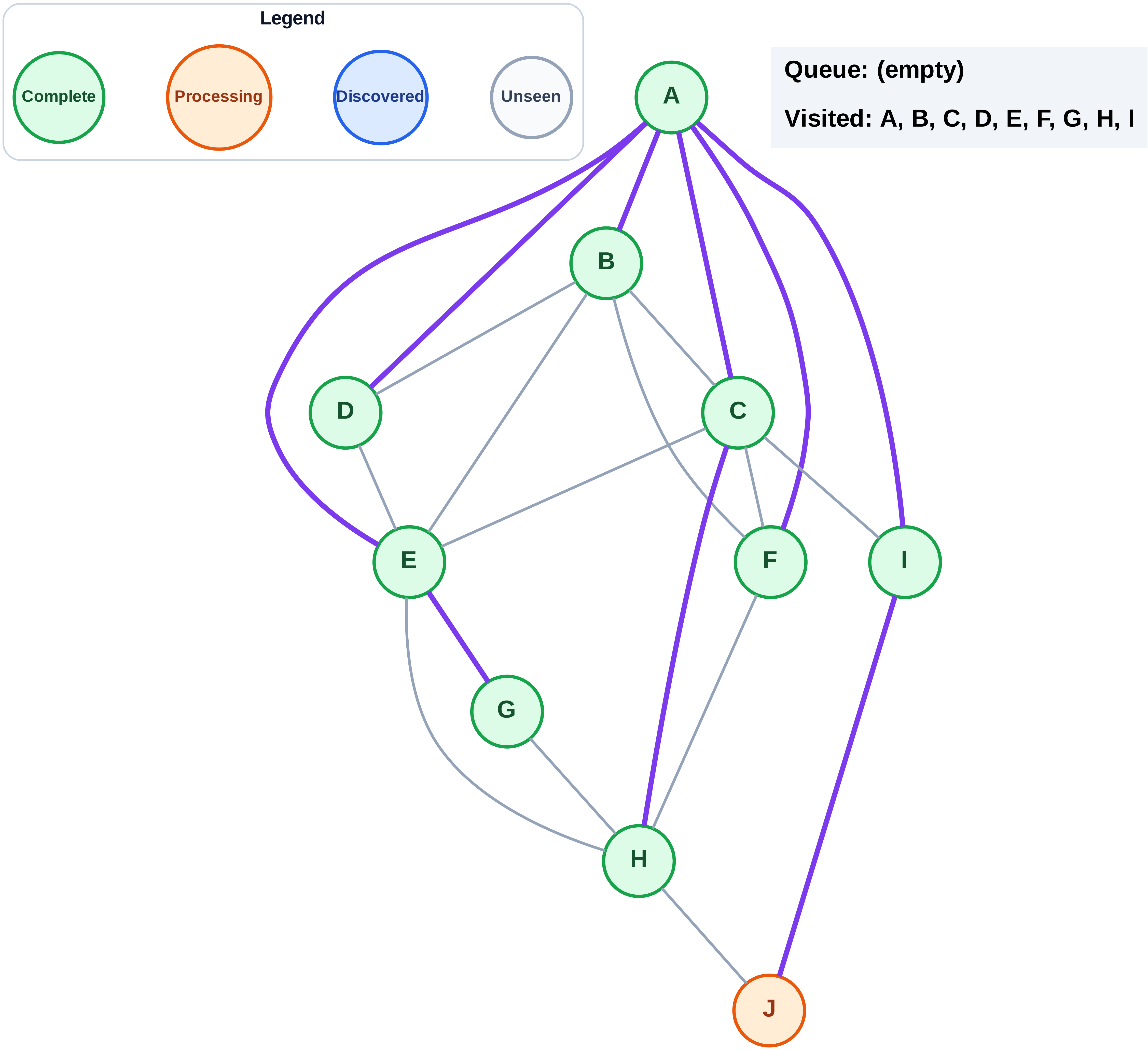




BFS Traversal
Step 19: No new neighbors from G
Legend
Complete
Processing
Discovered
Unseen
Queue: J
Visited: A, B, C, D, E, F, G, H, I
A
B
C
D
E
F
G
H
I
J

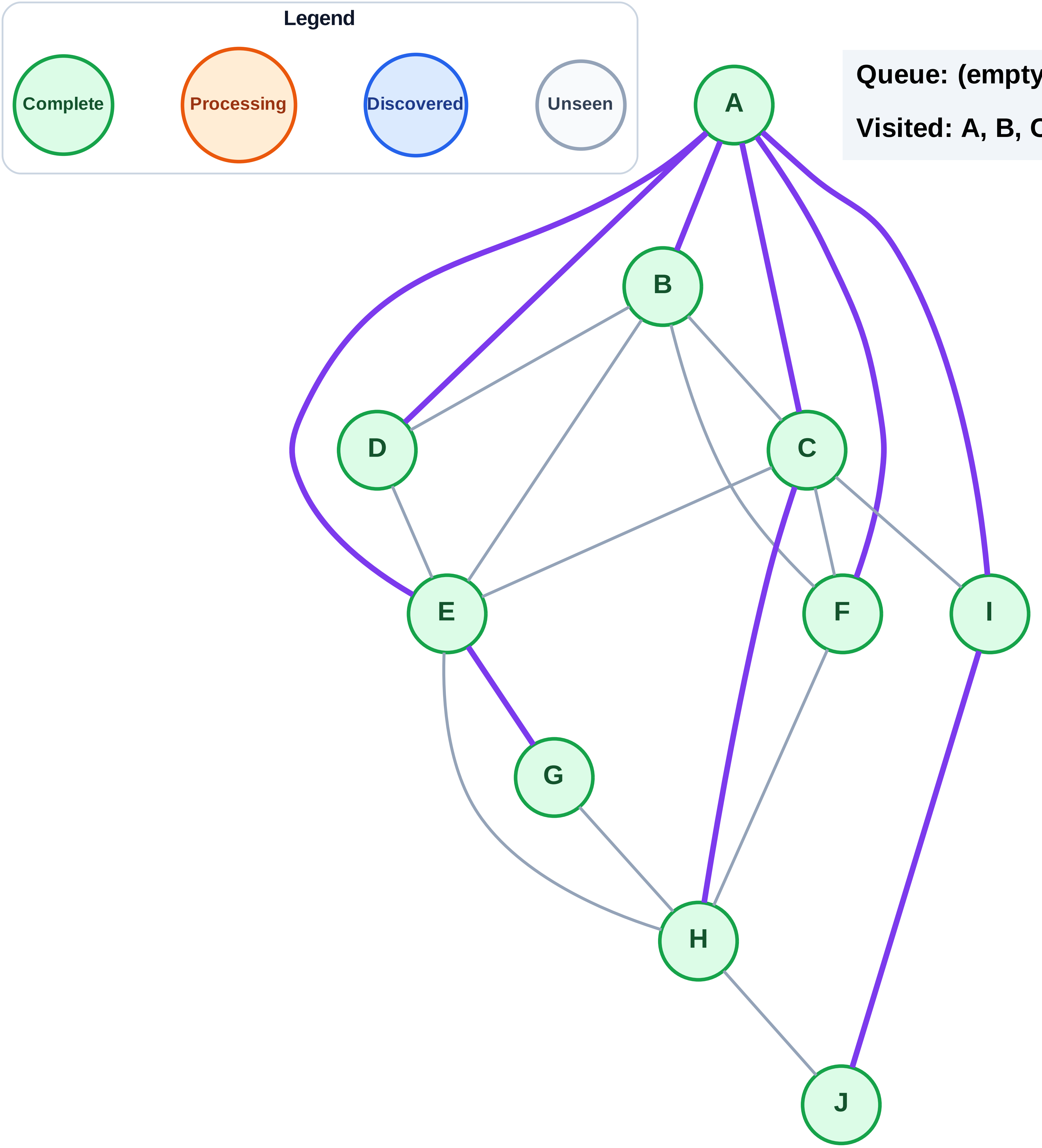
BFS Traversal

Step 20: Process node J



BFS Traversal

Step 21: No new neighbors from J

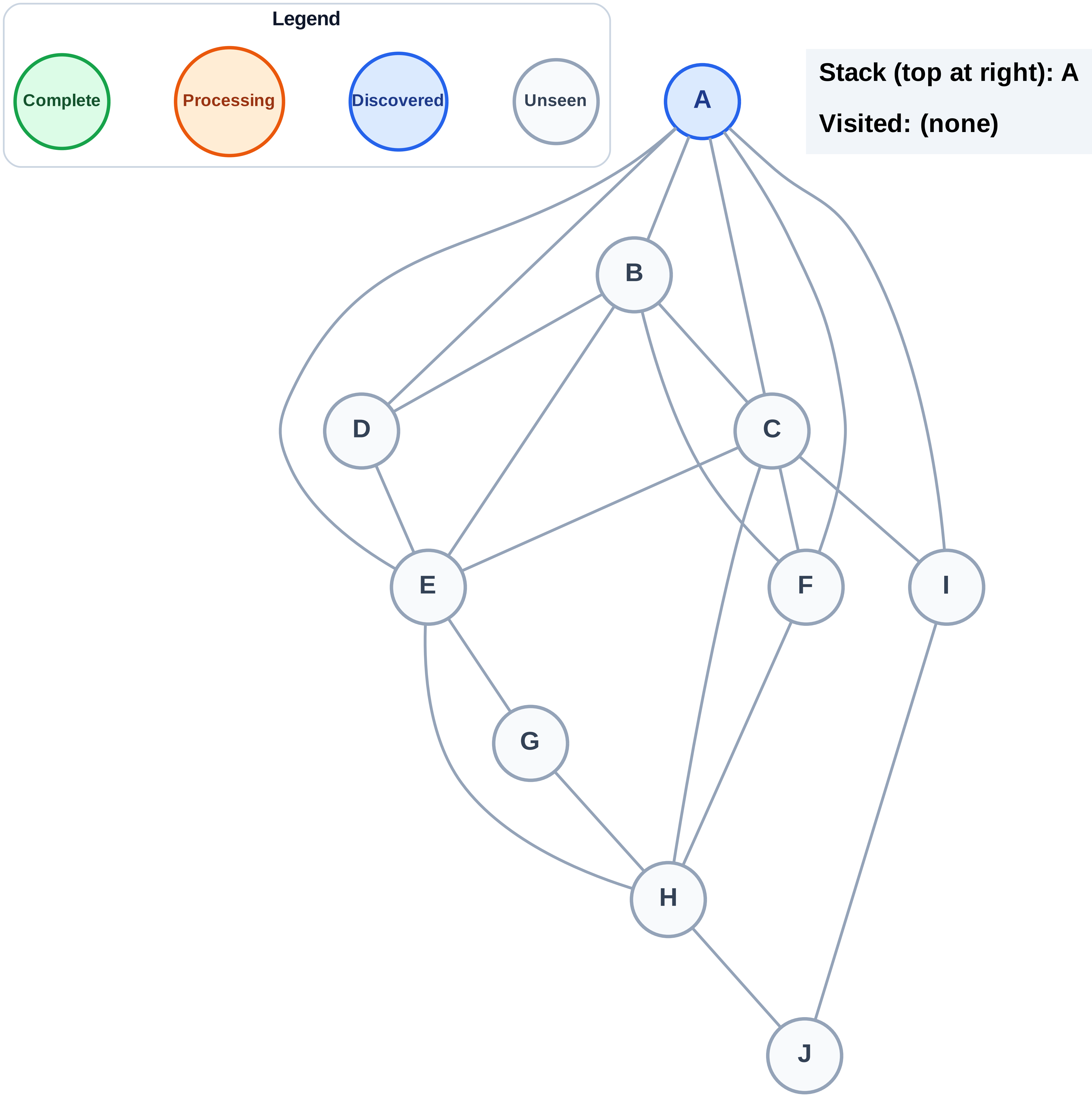


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

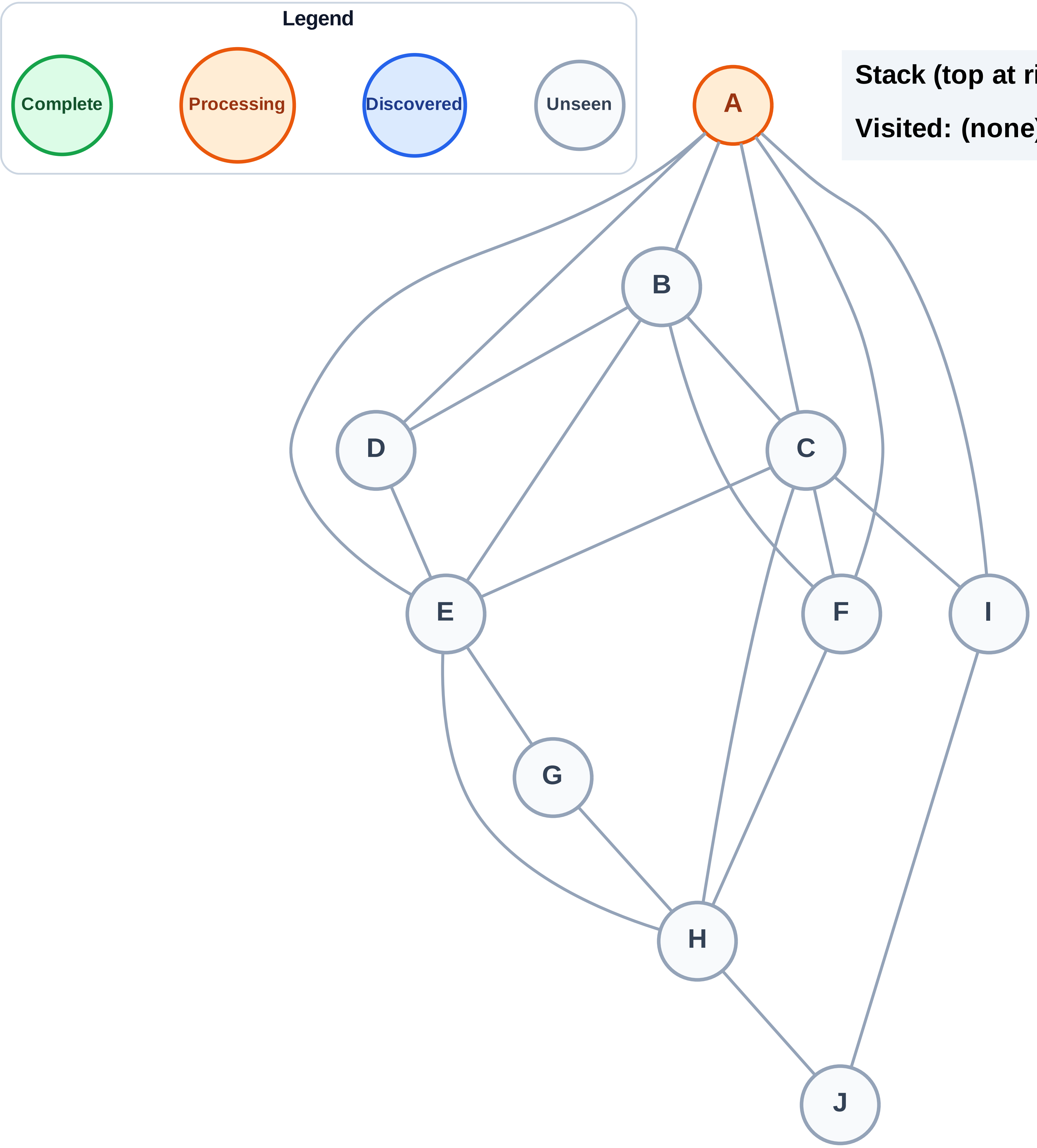
DFS Traversal

Step 1: Start at A



DFS Traversal

Step 2: Process node A

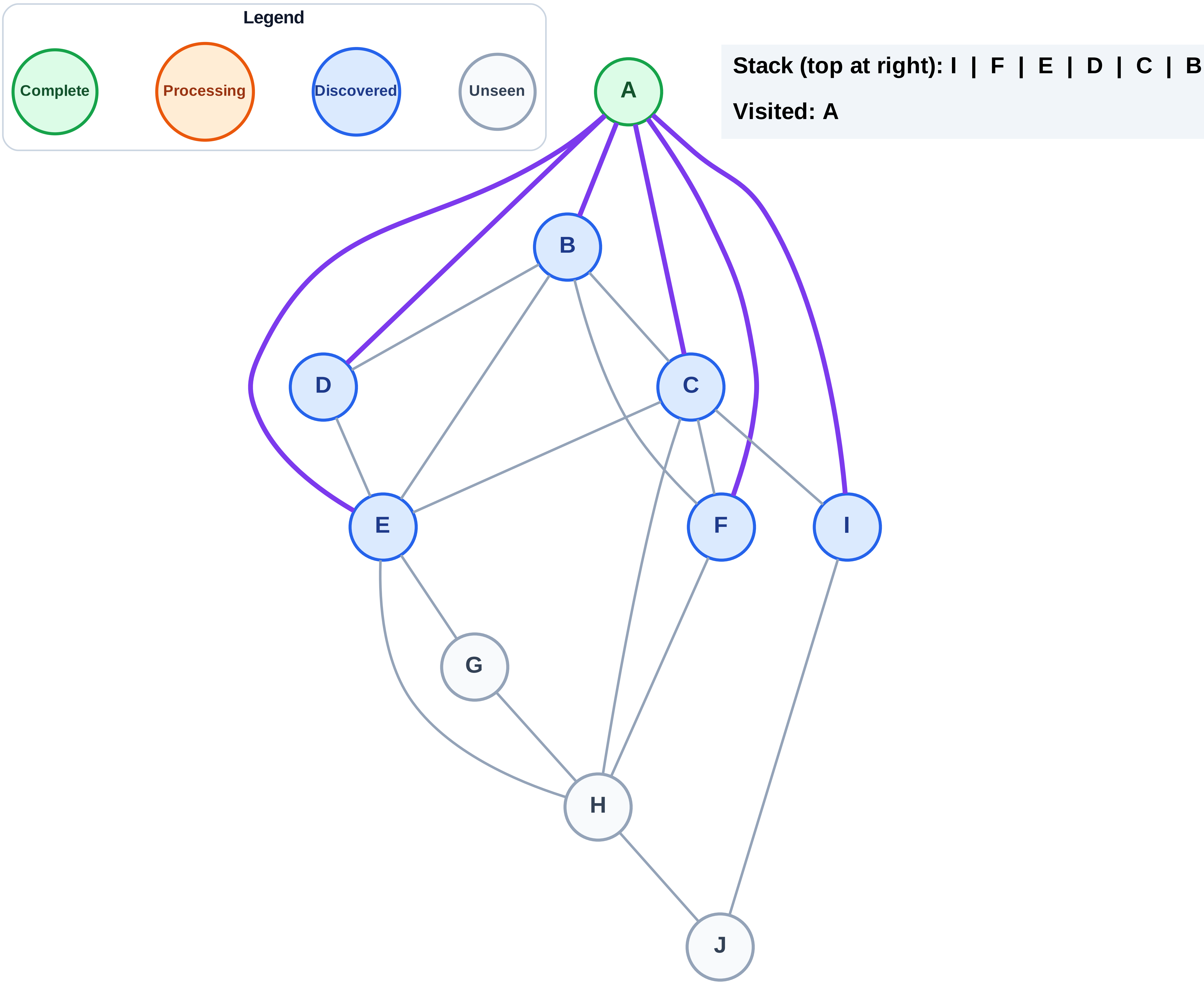


Stack (top at right): (empty)

Visited: (none)

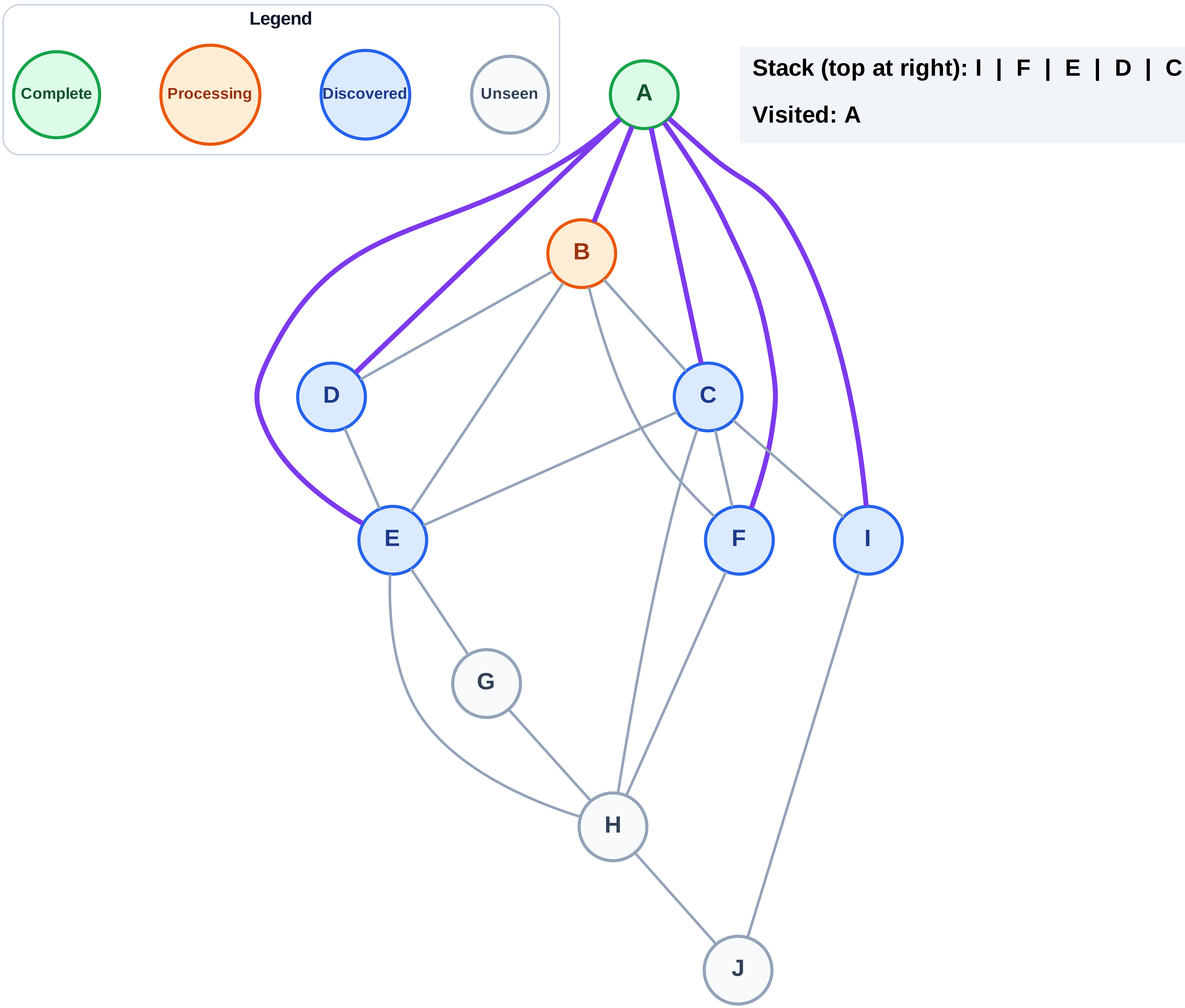
DFS Traversal

Step 3: Discover I, F, E, D, C, B from A



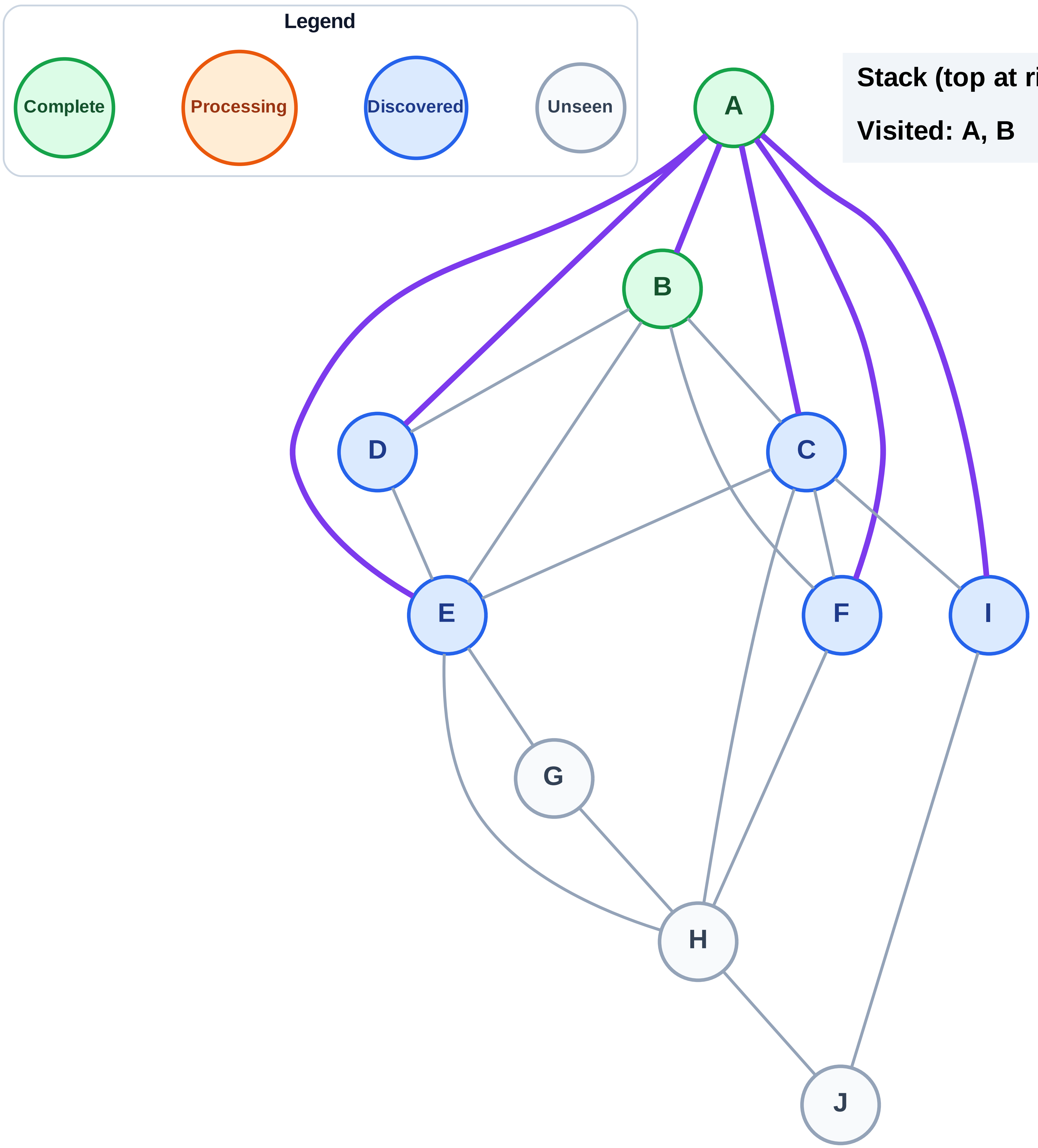
DFS Traversal

Step 4: Process node B



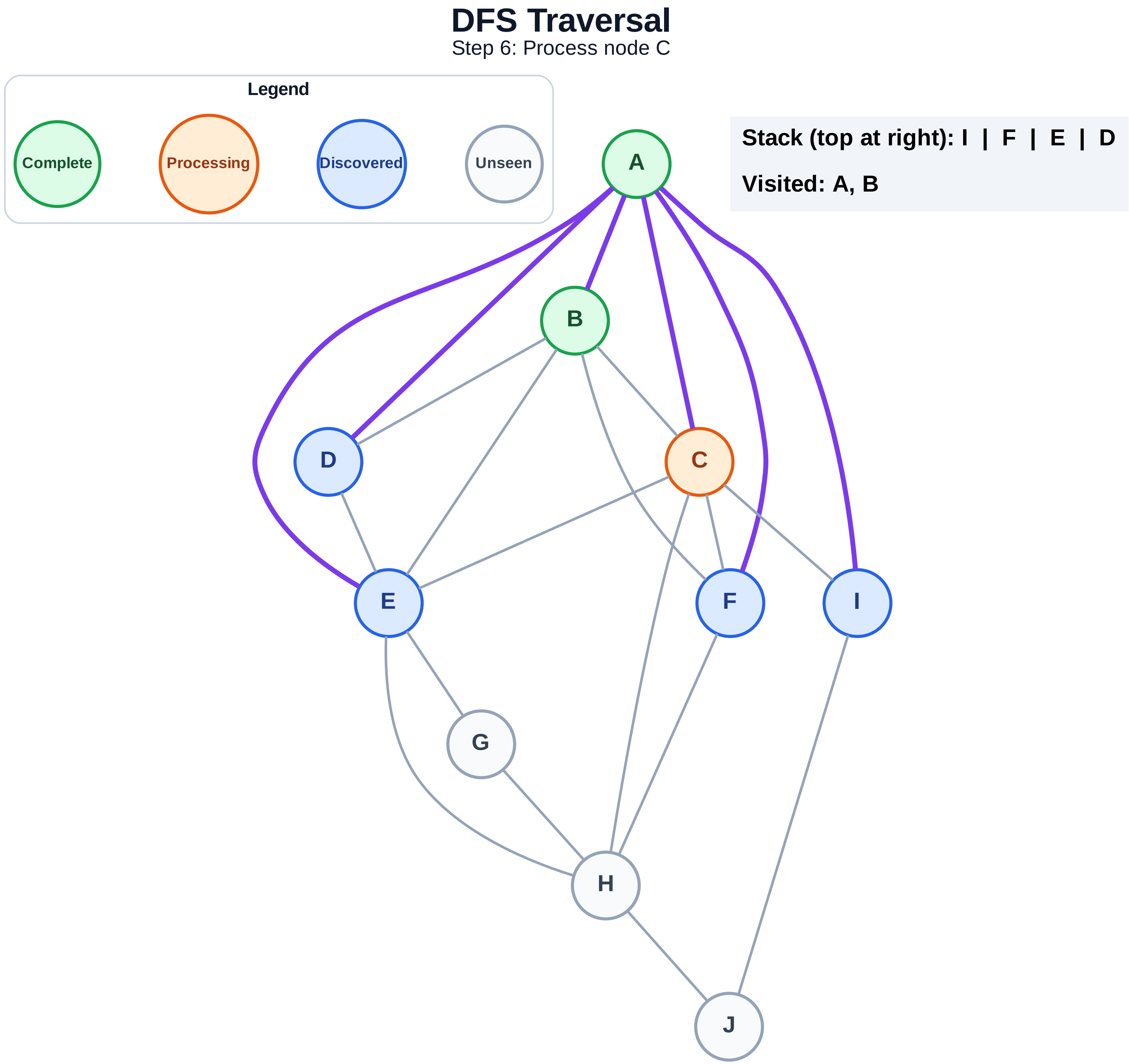
DFS Traversal

Step 5: No new neighbors from B



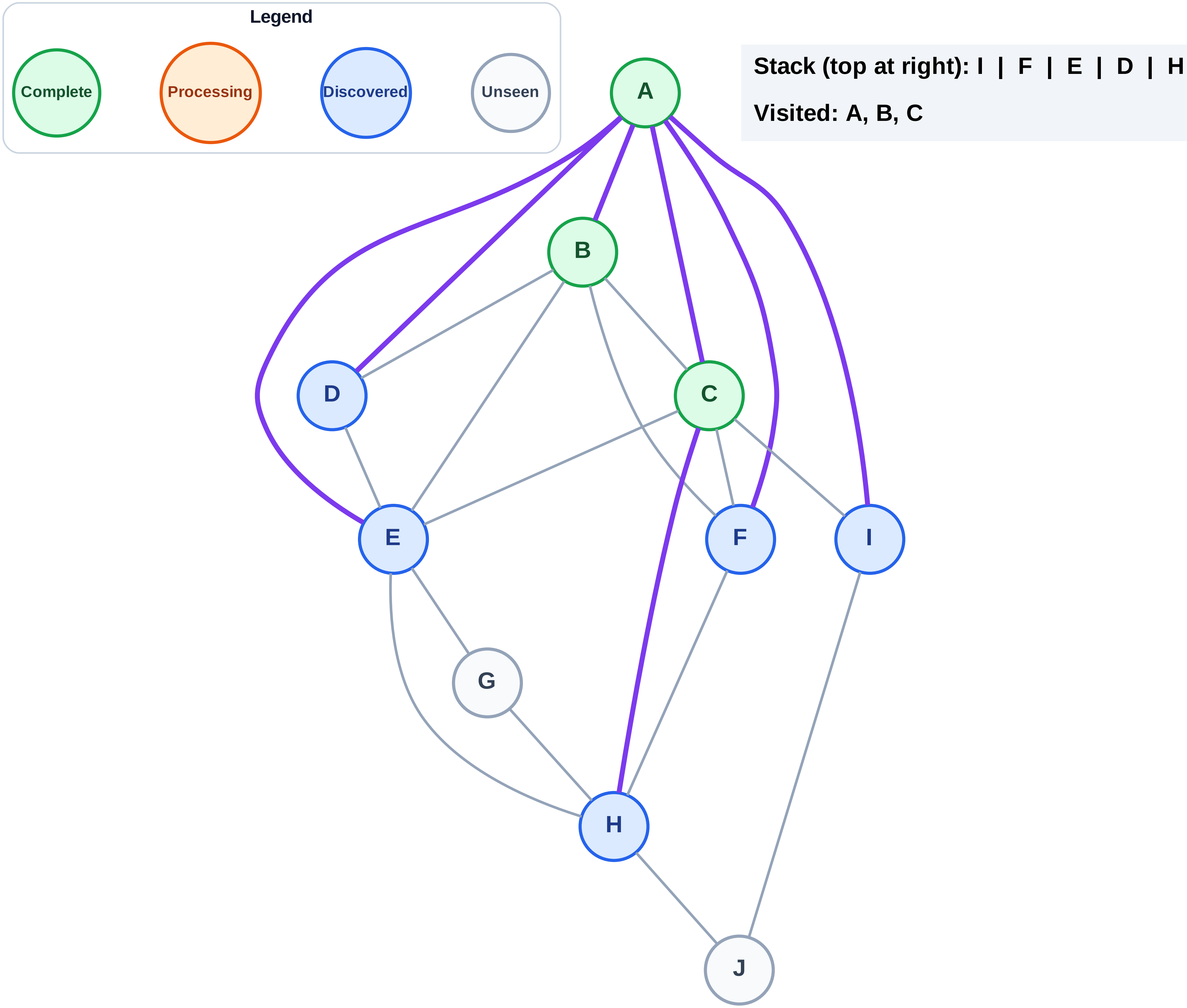
Stack (top at right): I | F | E | D | C

Visited: A, B


DFS Traversal
Step 6: Process node C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | F | E | D
Visited: A, B
A
B
C
D
E
F
G
H
I
J

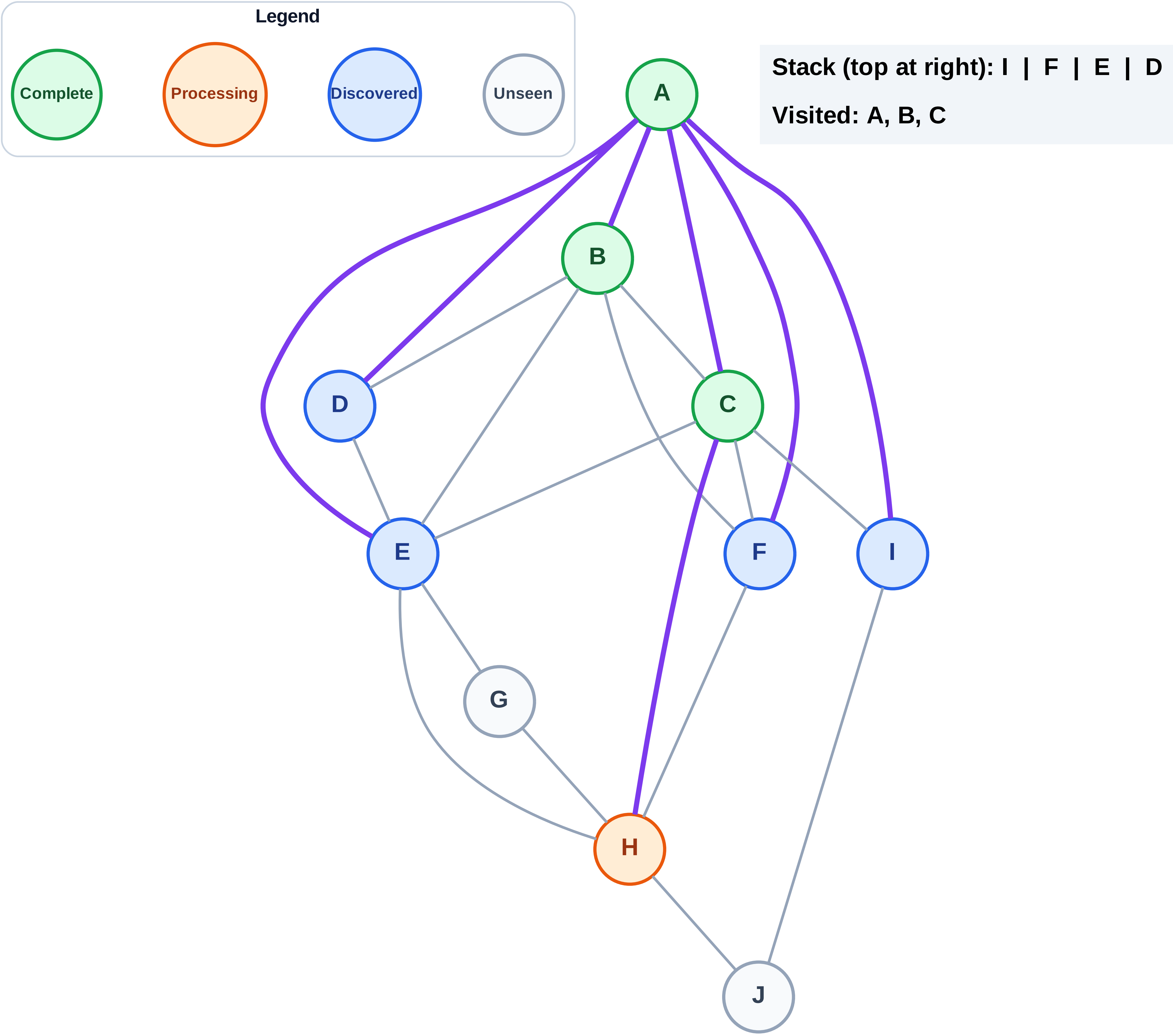
DFS Traversal

Step 7: Discover H from C



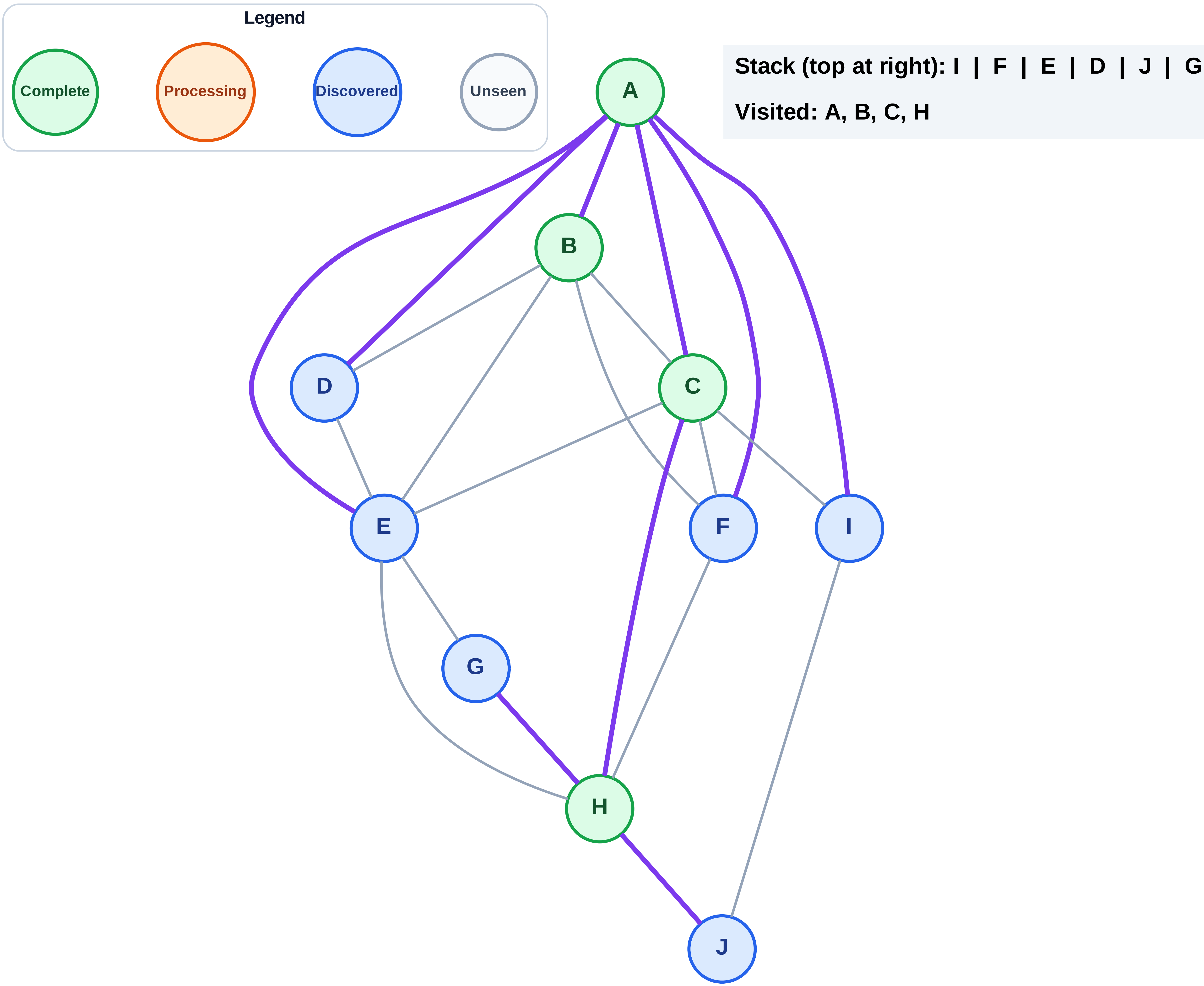
DFS Traversal

Step 8: Process node H



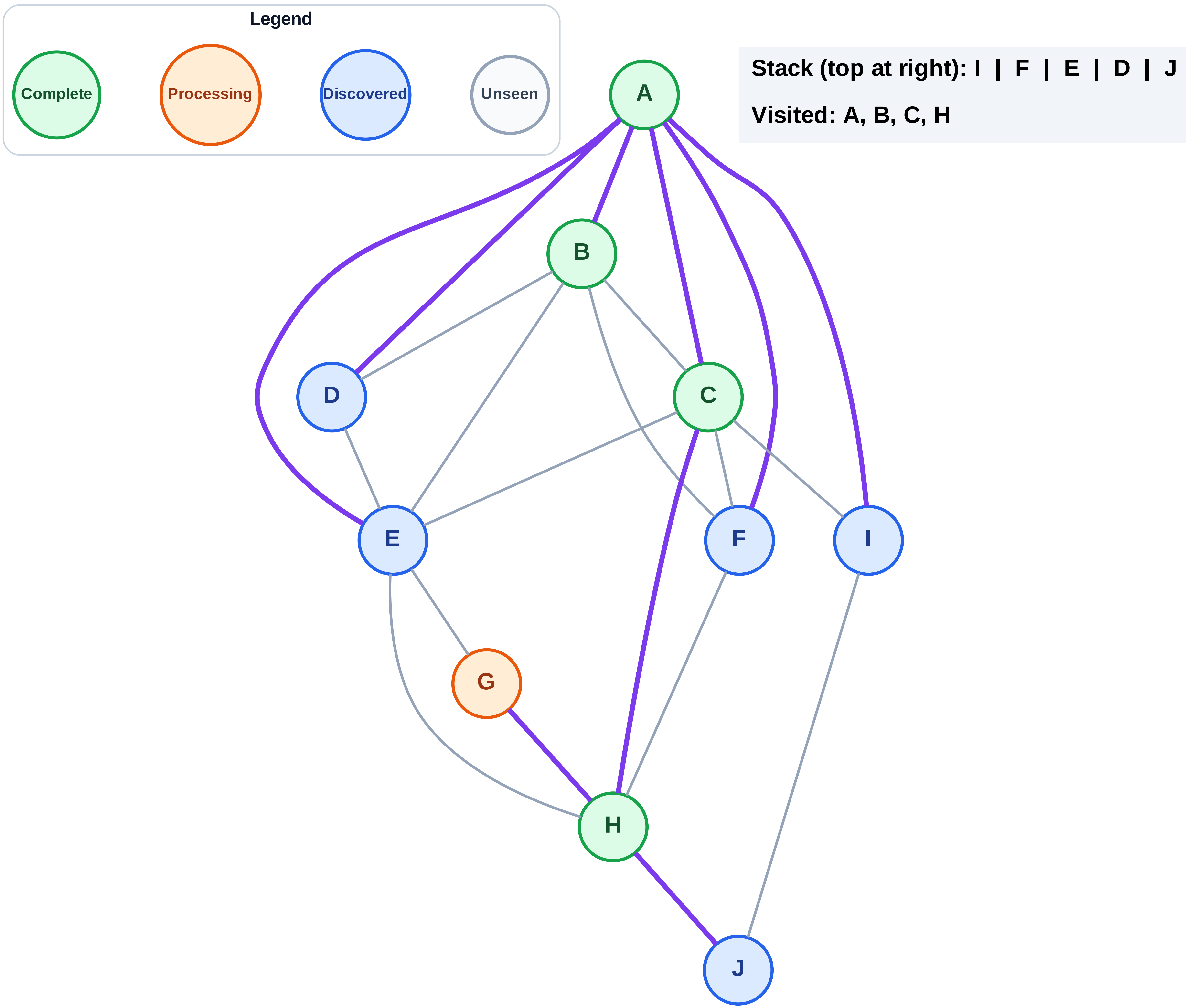
DFS Traversal

Step 9: Discover J, G from H



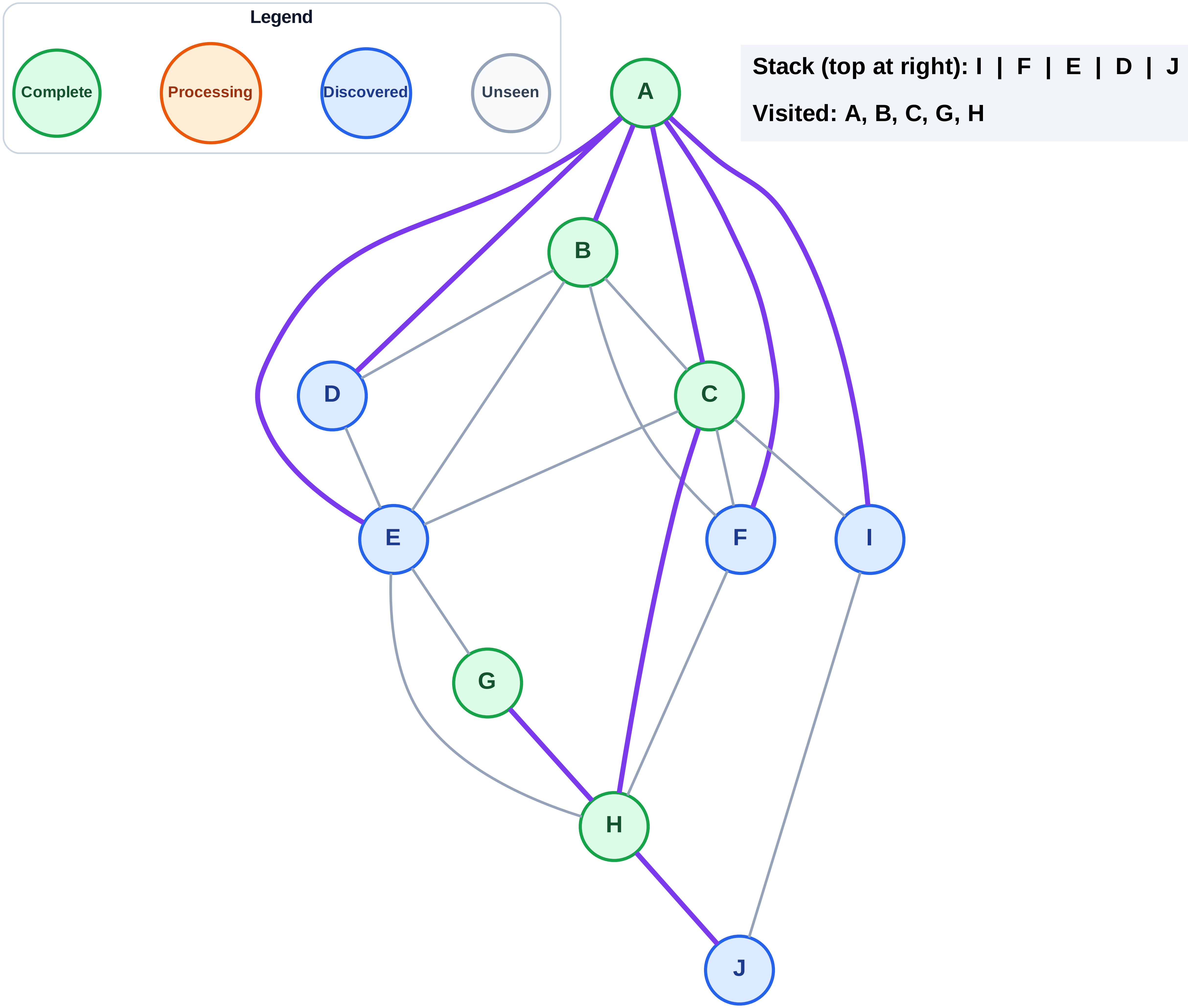
DFS Traversal

Step 10: Process node G



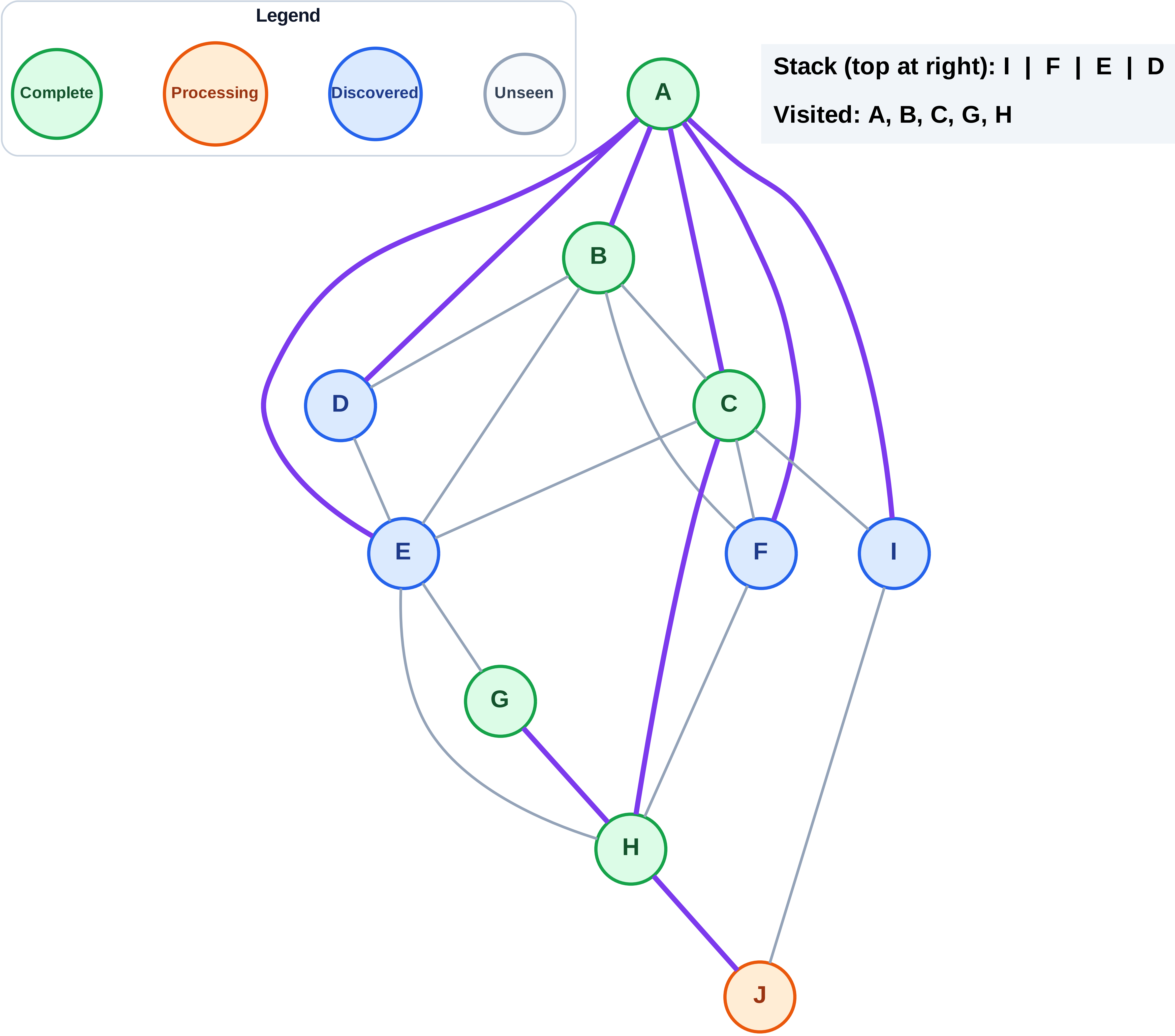
DFS Traversal

Step 11: No new neighbors from G



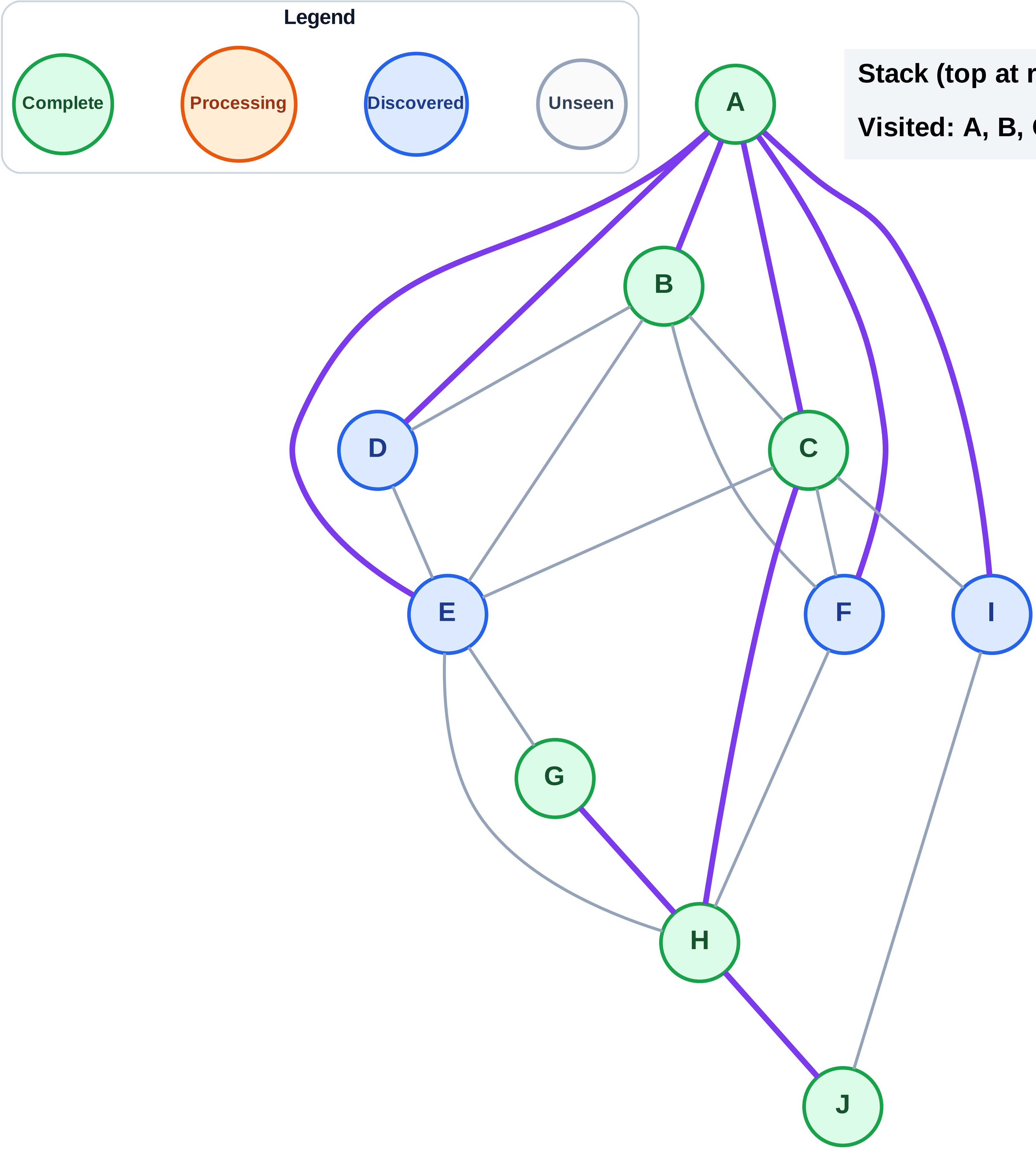
DFS Traversal

Step 12: Process node J



DFS Traversal

Step 13: No new neighbors from J

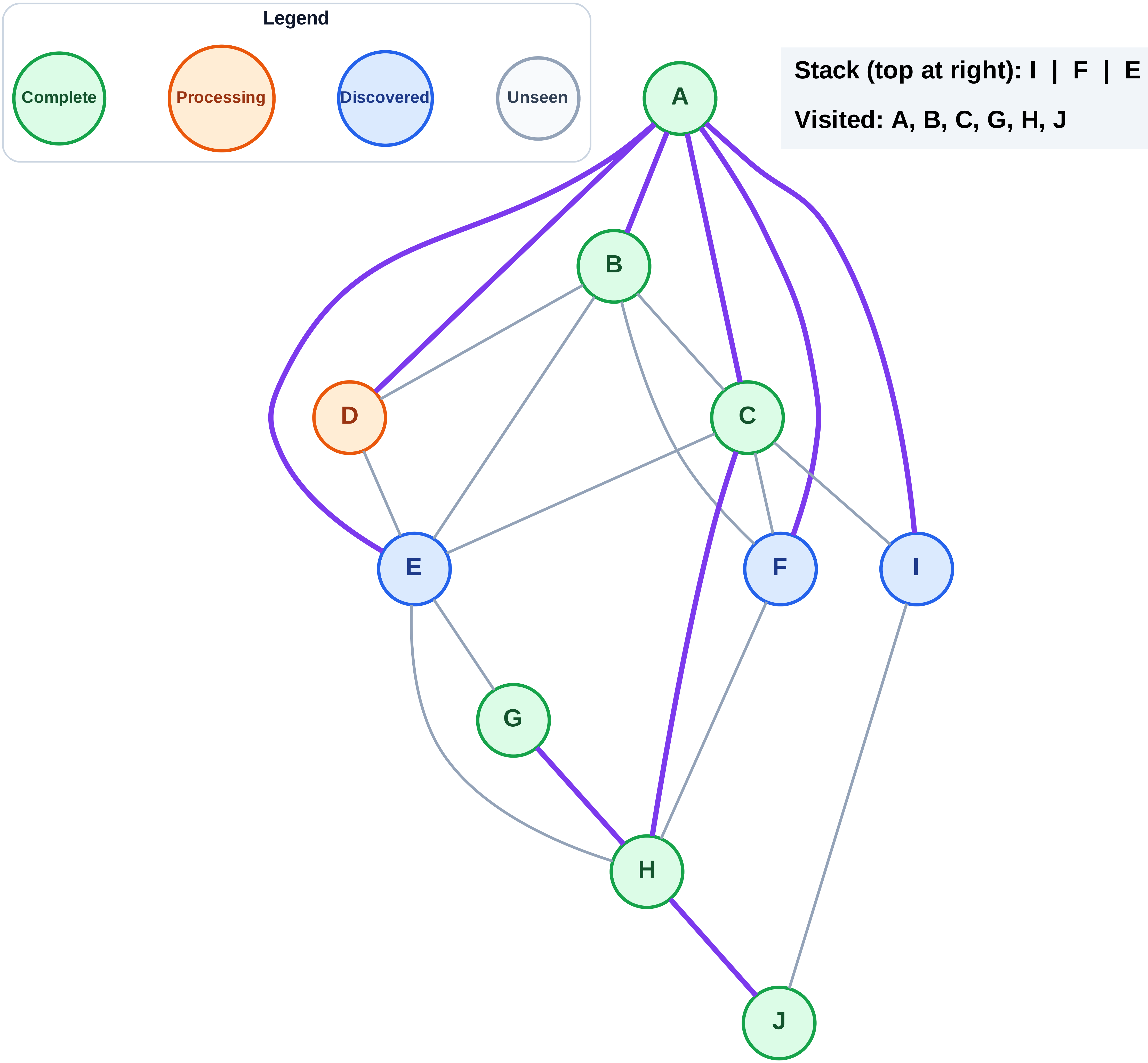


Stack (top at right): I | F | E | D

Visited: A, B, C, G, H, J

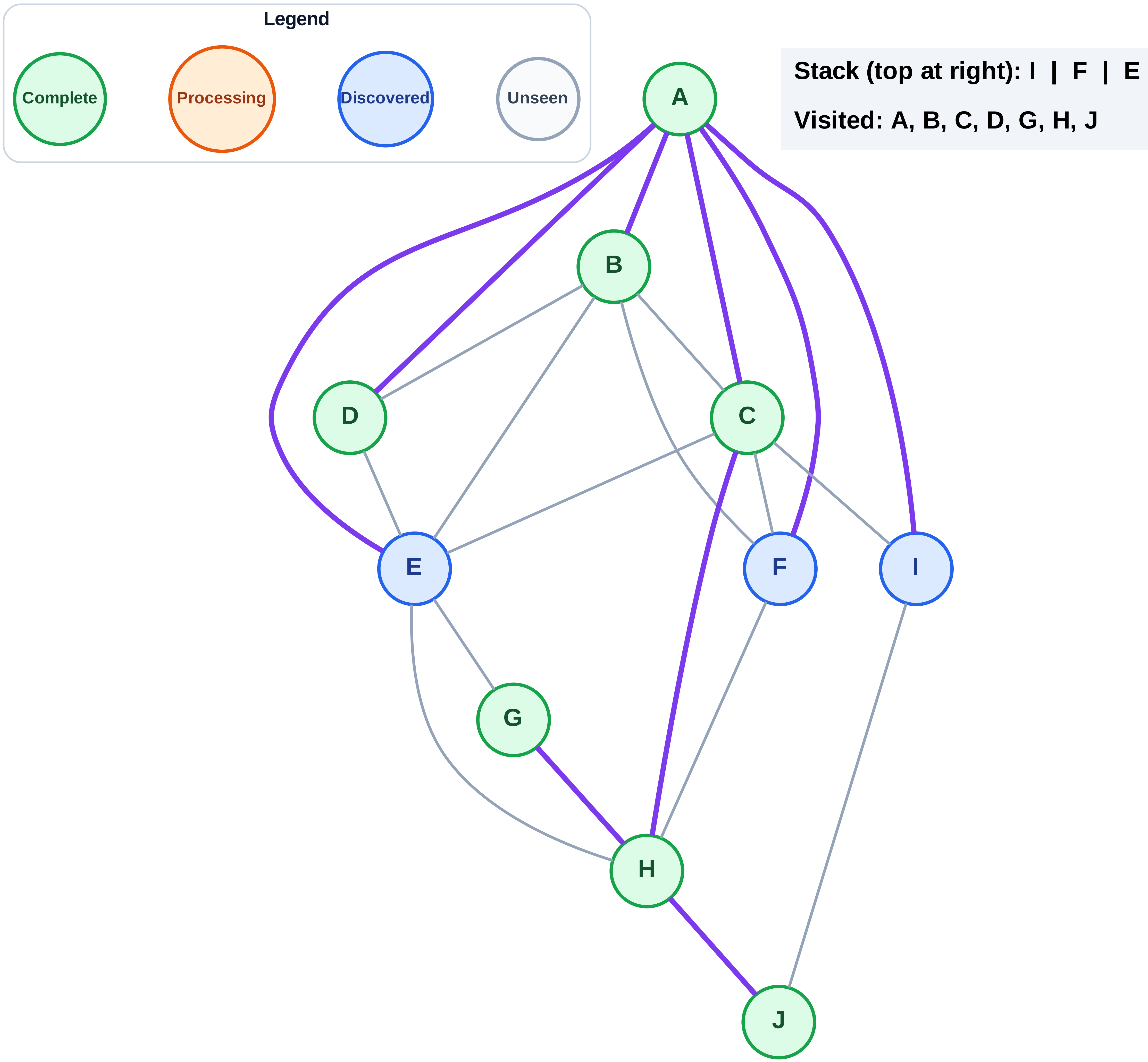
DFS Traversal

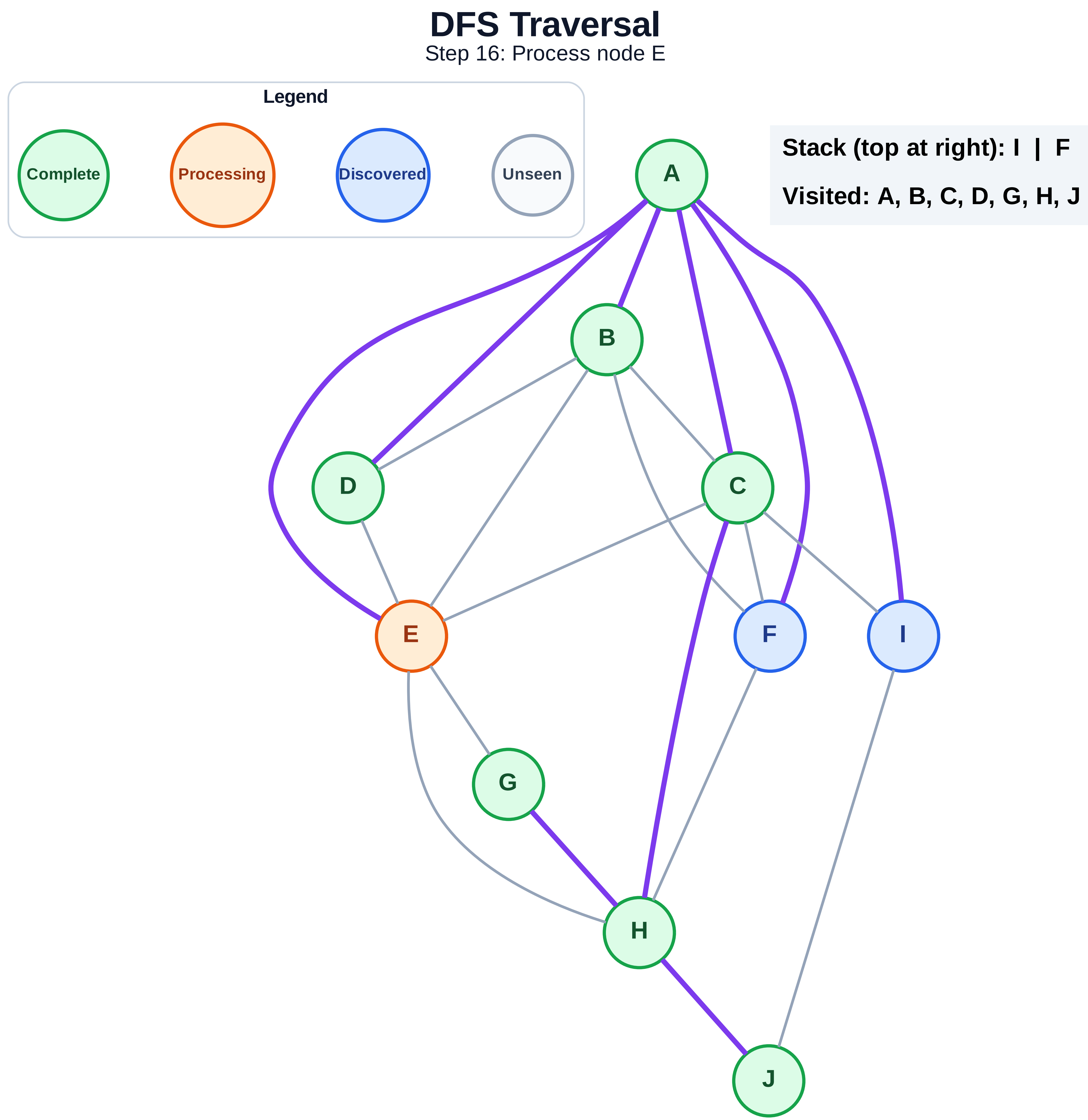
Step 14: Process node D



DFS Traversal

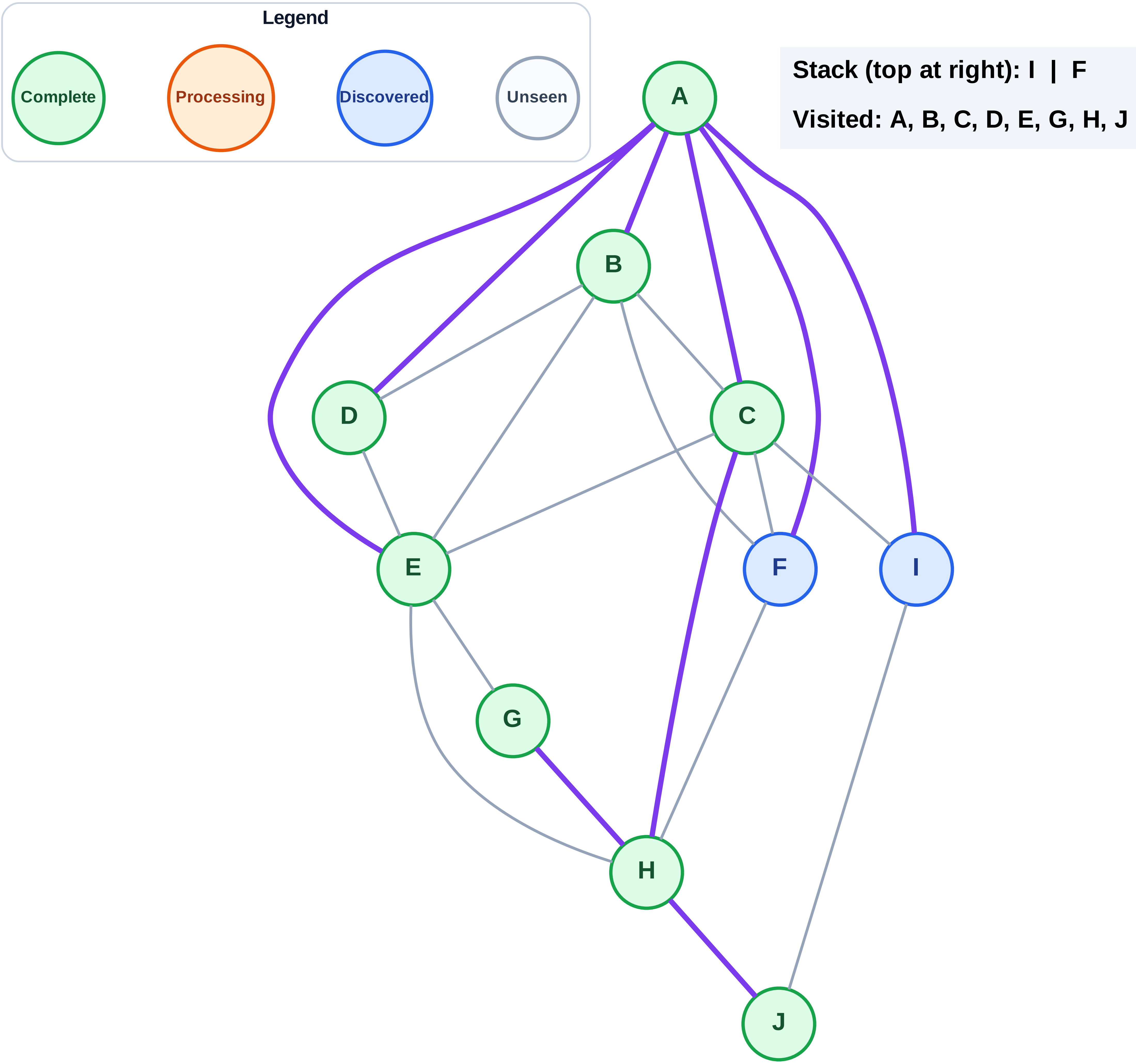
Step 15: No new neighbors from D




DFS Traversal
Step 16: Process node E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | F
Visited: A, B, C, D, G, H, J
A
B
C
D
E
F
G
H
I
J

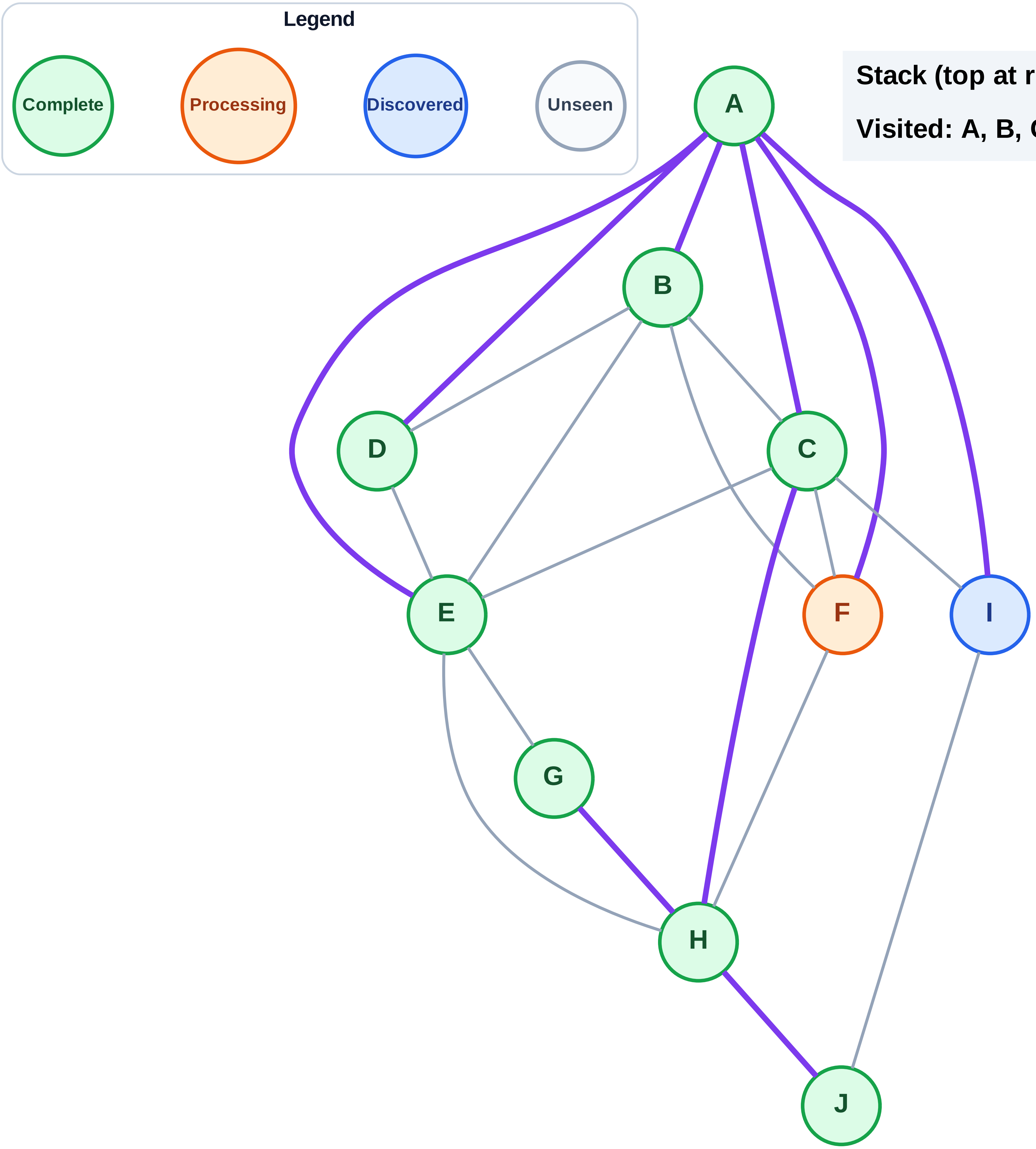
DFS Traversal

Step 17: No new neighbors from E



DFS Traversal

Step 18: Process node F

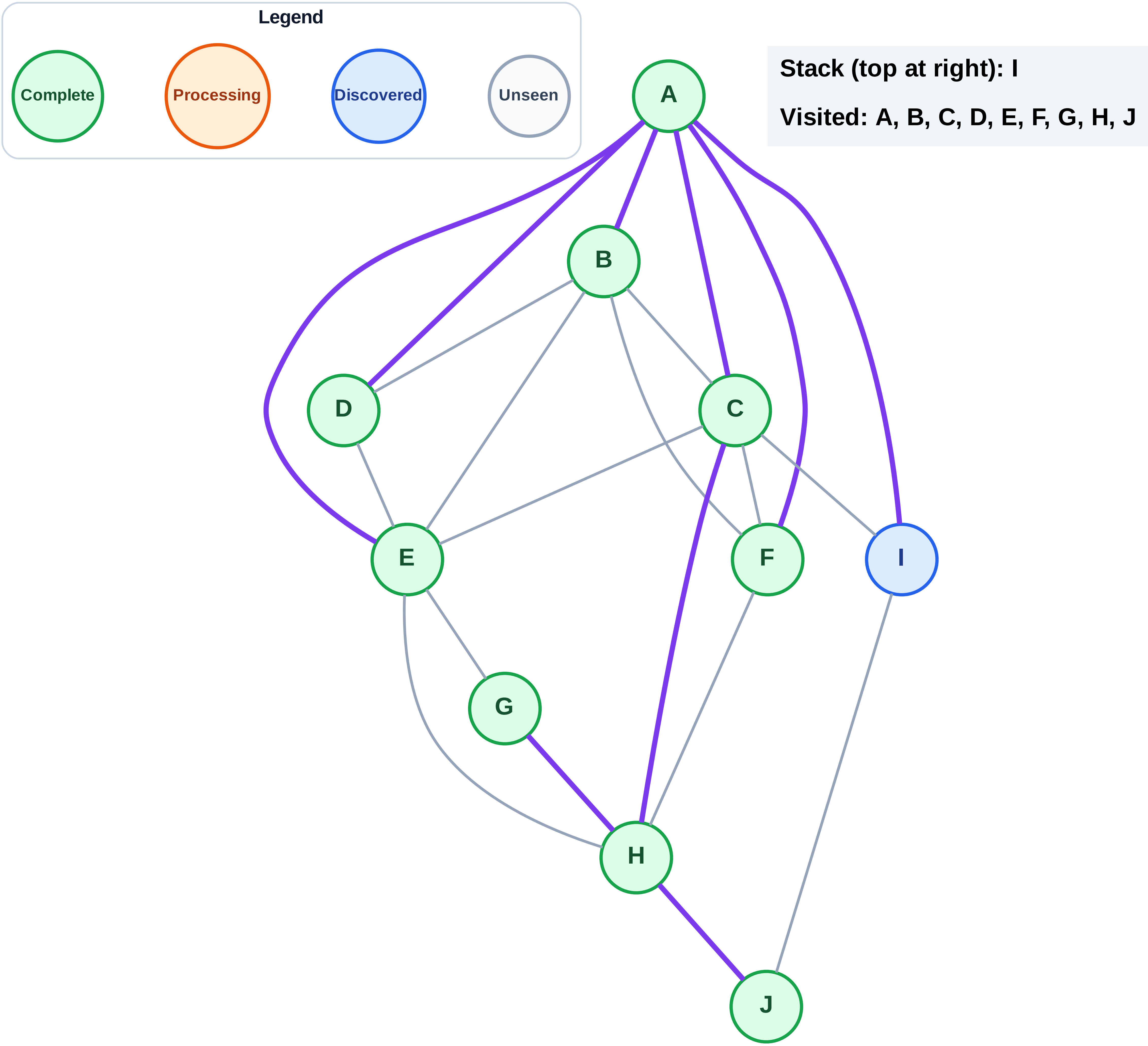


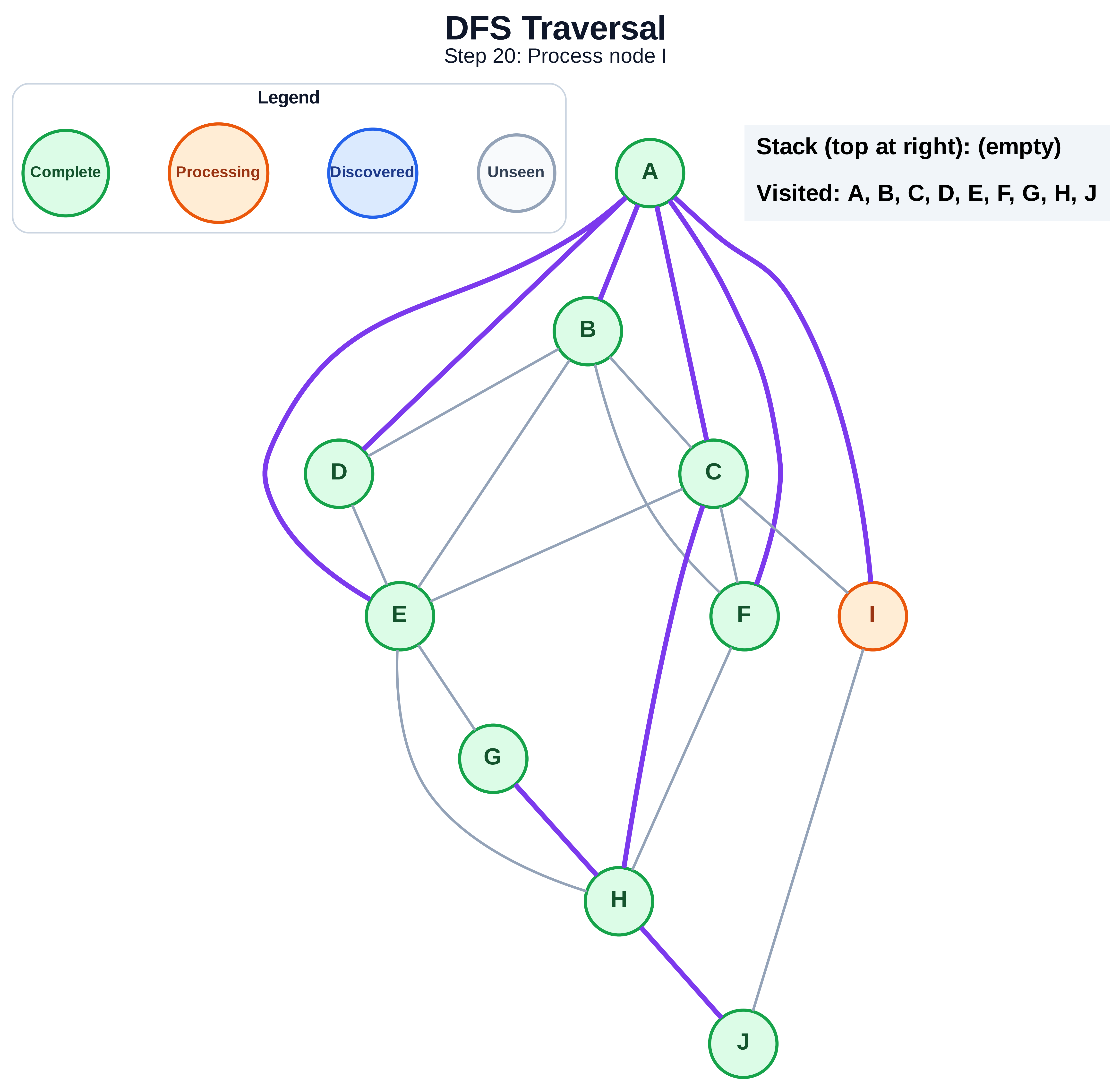
Stack (top at right): I

Visited: A, B, C, D, E, G, H, J

DFS Traversal

Step 19: No new neighbors from F




DFS Traversal
Step 20: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 21: No new neighbors from I